

1

In this chapter, we will learn about

- the photoelectric effect
- the limitations of the wave model of light in explaining the photoelectric effect
- Einstein's interpretation of the photoelectric effect

Particle Nature of Light

1.1 The puzzling photoelectric effect

Flipped classroom
Watch a video and answer the questions.

When talking about genius, many people will think of Albert Einstein (Fig 1.1a). He is best known for developing the *theory of relativity* and deducing the equation $E = mc^2$. He received the Nobel prize in physics in 1921, not for the above achievements, but for his discovery of the law of the **photoelectric effect**. What is so special about this effect?

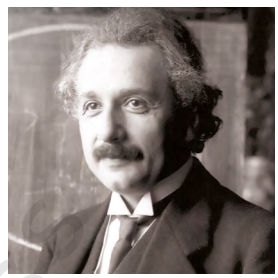


Fig 1.1a Albert Einstein.

1 Studying photoelectric effect

a What is photoelectric effect?

Cross-link
You learned gold leaf electroscope in Book 4 Ch 1.1.

When a negatively charged metal plate is put on the metal cap of a gold leaf electroscope, the entire conducting part, including the metal cap, the metal support and the gold leaf, become negatively charged. Therefore, the gold leaf deflects upwards (Fig 1.1b(i)).

When ultra-violet radiation shines on the metal plate, the gold leaf drops quickly. This shows that the negative charges on the metal plate decrease. In fact, electrons are emitted from the metal plate when certain electromagnetic radiation shines on it (Fig 1.1b(ii)). This is the photoelectric effect. The electrons emitted are called **photoelectrons**.

This effect can make a negatively charged material neutral, and a neutral material positively charged.

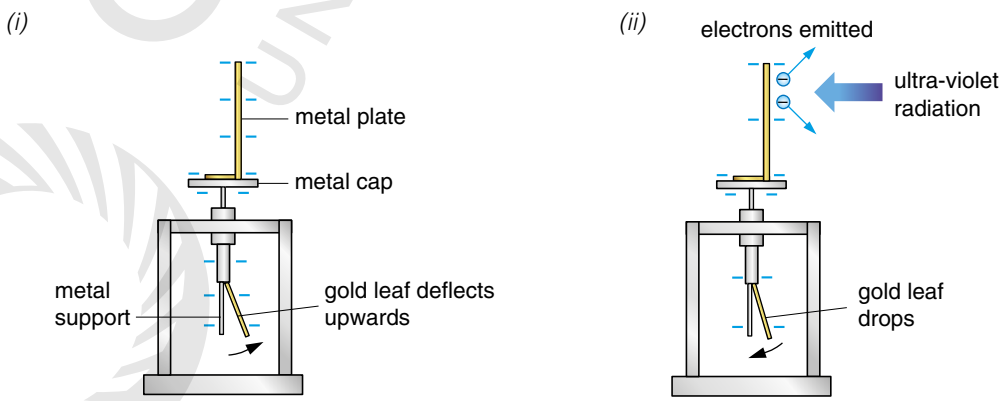


Fig 1.1b Using a gold leaf electroscope to demonstrate the photoelectric effect.

→ This video demonstrates Expt 1a.

Experiment 1a
Photoelectric effect (p.138)
Video

The photoelectric effect is the emission of electrons from the surface of a material exposed to electromagnetic radiation.

theory of relativity 相對論　photoelectric effect 光電效應　photoelectron 光電子

DSE goal

Describe photoelectric effect experiment and its results.

DSE exam

15(2)Q2.5, 19(2)Q2.2

b Photoelectric effect experiment

A **photocell** (Fig 1.1c) is an evacuated tube containing two electrodes, the *cathode* (a metal plate) and the *anode*. It can be used to study the photoelectric effect.

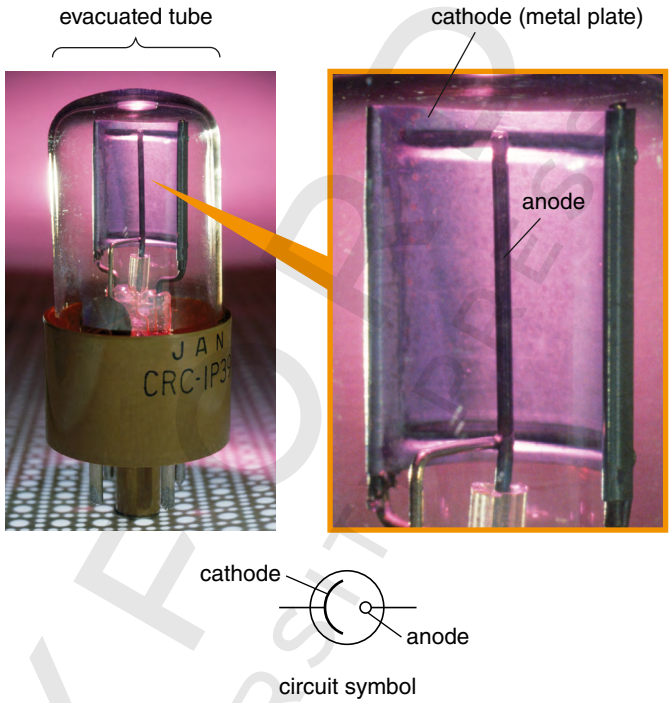


Fig 1.1c *A photocell and its circuit symbol.*

Figure 1.1d shows the set-up of the photoelectric effect experiment. The photocell is connected to an ammeter and a power supply whose output voltage can be varied in both magnitude and direction.

At first, there is no current flowing in the circuit as electrons cannot get through the space between the cathode and the anode. To study the photoelectric effect, light is directed onto the cathode. If an electron on the cathode can absorb enough energy from the light, it will be emitted (Fig 1.1e).

In physics, we often refer to radiation from any part of the electromagnetic spectrum as 'light'.

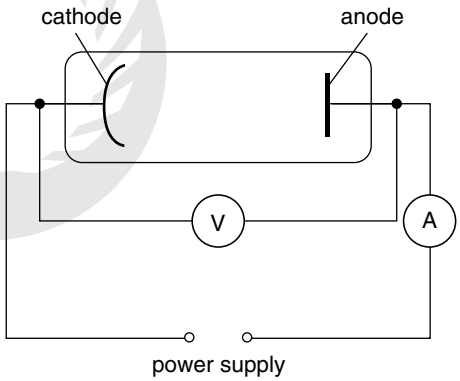


Fig 1.1d *Experiment set-up used to study the photoelectric effect.*

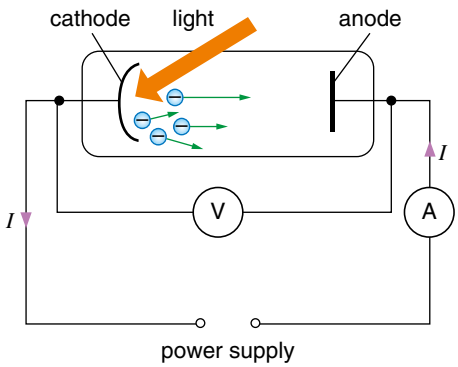


Fig 1.1e *Photoelectrons are emitted from the cathode.*

Know more

Heinrich Hertz (1857–1894)



The photoelectric effect was first observed and reported by the German scientist *Heinrich Hertz* in 1887. Hertz was also the first scientist to conclusively prove the existence of electromagnetic waves experimentally.

photocell 光電池 cathode 陰極 anode 陽極 Heinrich Hertz 赫茲

Photoelectrons are emitted from the cathode in **different directions** (Fig 1.1f). Some of them reach the anode. The flow of electrons from the cathode to the anode forms a current, called the **photoelectric current**, and is measured by the ammeter. The size of this current is proportional to the number of photoelectrons **reaching the anode** per second.

The photoelectric current in the photocell flows from the anode to the cathode, opposite to the direction of flow of the photoelectrons.

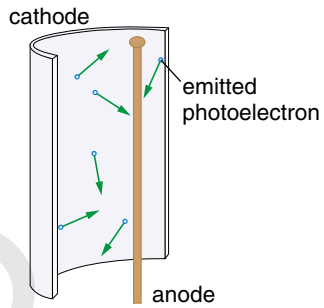


Fig 1.1f Photoelectrons emitted in different directions.

The p.d. V across the cathode and the anode can be varied continuously by changing the output voltage of the power supply. Figure 1.1g shows a graph of the photoelectric current I against V. V is taken as positive when the anode is connected to the positive terminal of the power supply.

May remind Ss that $V = IR$ cannot be applied to photocells.

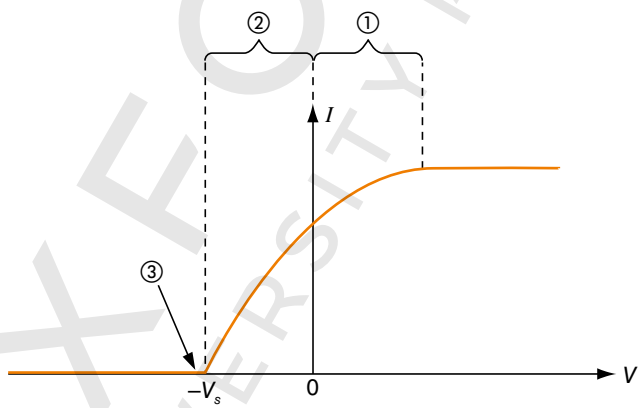


Fig 1.1g Photoelectric current I against p.d. V across the cathode and the anode.

The following can be observed from this graph:

① When V is positive, the photoelectrons are promoted to reach the anode (Fig 1.1h). When V gradually increases, more and more photoelectrons reach the anode, so I increases. When V is high enough, all photoelectrons emitted reach the anode and I reaches a constant value called the **saturated current**.

The saturated current is equal to the rate of emission of charges from the cathode and, therefore, indicates the rate of emission of photoelectrons. For example, a saturated current of 1 A means that 1 C of charge (6×10^{18} electrons) is emitted per second.

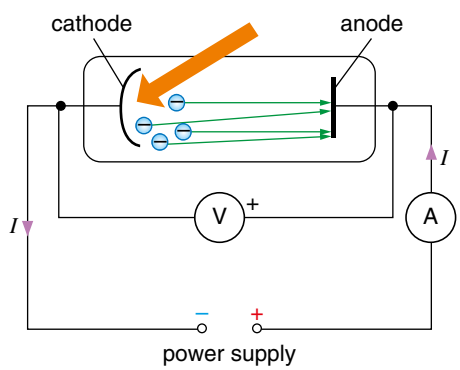


Fig 1.1h Photoelectrons are promoted to reach the anode when V is positive.

photoelectric current 光電流　　saturated current 飽和電流

② When V becomes negative, the photoelectrons are hindered from reaching the anode. When V becomes more negative, I decreases. This is because the photoelectrons are emitted with a range of kinetic energy. Fewer and fewer photoelectrons have enough kinetic energy to overcome the p.d. to reach the anode (Fig 1.1i).

What will happen to the balls if height h increases? How is this analogous to the situation in the photocell?

See p.6 for the answers.

Figure 1.1j shows an analogy that helps understand the result. Only balls (photoelectrons) with enough kinetic energy can pass the inclined plane.

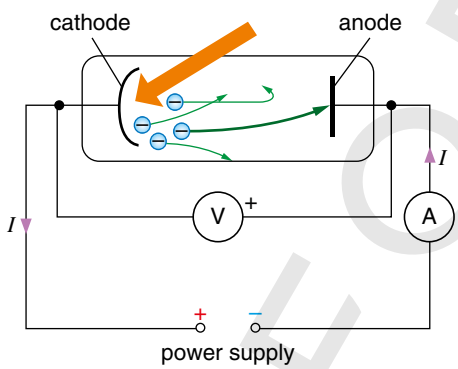


Fig 1.1i *When V is negative, photoelectrons with low KE cannot overcome the p.d. to reach the anode.*

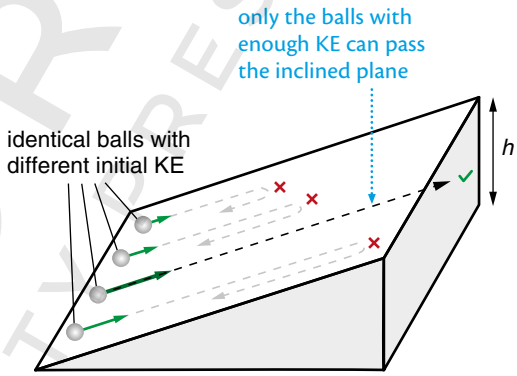


Fig 1.1j *An analogy to the photoelectrons moving towards the anode.*

③ I drops to zero when V is just enough to stop the **most energetic photoelectrons** from reaching the anode (Fig 1.1k). The **magnitude** of this p.d. is called the **stopping potential** V_s.

The stopping potential is the magnitude of the potential difference that is just enough to stop the photoelectrons with the maximum kinetic energy from reaching the anode.

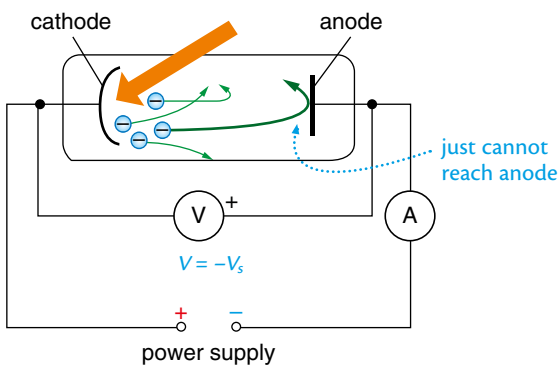


Fig 1.1k *V is just enough to stop the most energetic photoelectrons from reaching the anode.*

stopping potential 遏止電勢

The magnitude of the charge of a photoelectron is e. The maximum kinetic energy of the photoelectrons emitted from the cathode is K_{max}. When a photoelectron with initial kinetic energy K_{max} is just stopped by the stopping potential V_s, its gain in electric potential energy is eV_s (Fig 1.1l).

Cross-link
Refer to Book 5 p.99.

Electric potential energy is also introduced in Book 4 Ch 1 p.54 *Know more*.

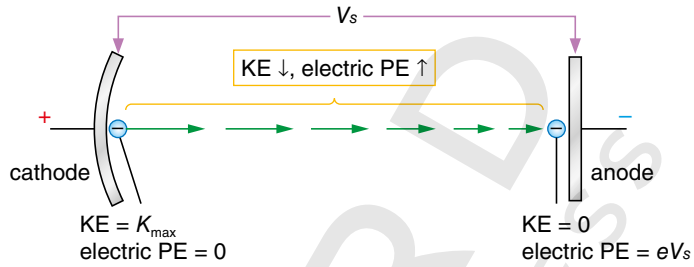


Fig 1.1l *Energy change of the photoelectron with maximum KE.*

By the law of conservation of energy,

max. KE of photoelectron (eV) $K_{max} = eV_s$ magnitude of **charge** of electron; **stopping potential** (V)

1 eV = $e \times 1$ V

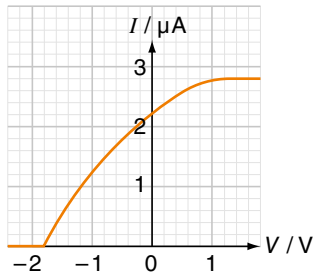
We can obtain K_{max} in J by substituting $e = 1.60 \times 10^{-19}$ C into the equation.

Since e is a constant, the value of the stopping potential V_s is a measure of the maximum kinetic energy K_{max} of the photoelectrons emitted from the cathode.

In the analogy in Figure 1.1j, when h ↑, the number of balls passing the inclined plane ↓. When h is increased to h_s, the ball with max. KE (K_{max}) just cannot pass the inclined plane. When it reaches the top of the plane, all its KE is converted to PE (mgh_s). We have $K_{max} = mgh_s$. Hence, h_s is a measure of the max. KE of the balls.

Example 1 Photoelectric effect experiment

In the photoelectric effect experiment, the graph of the photoelectric current I against the potential difference V across the photocell is as shown.



(a) What is the maximum KE (in eV) of the photoelectrons emitted?

(b) Estimate the number of photoelectrons emitted per second.

Solution

(a) $K_{max} = eV_s = e \times 1.8$ V $= 1.8$ eV

(b) Saturated current $= 2.8 \times 10^{-6}$ A

In other words, 2.8×10^{-6} C of charge is emitted in each second.

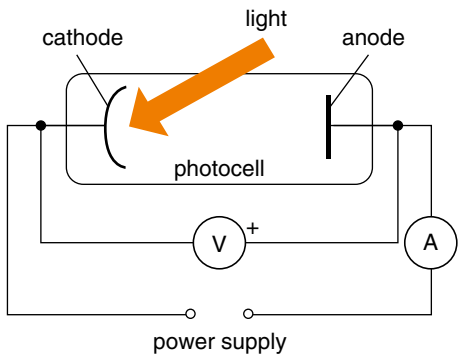
Number of photoelectrons emitted per second

$$= \frac{Q}{e} = \frac{2.8 \times 10^{-6}}{1.60 \times 10^{-19}} = 1.75 \times 10^{13}$$

Quick check 1

→ E-version in OUP Exercise Platform is also available. Google Forms Microsoft Forms

(For Q1–2.) Consider the set-up shown below. Light falls on the cathode of the photocell and photoelectric effect occurs. The voltage V across the photocell is taken as positive when the anode is connected to the positive terminal of the power supply.



1 Determine whether the following statements are true.

(a) Photoelectrons emitted from the cathode can always reach the anode. (T/F)

(b) When V is positive, the photoelectrons emitted from the cathode are promoted to reach the anode. (T/F)

(c) No photoelectrons are emitted when V is zero. (T/F)

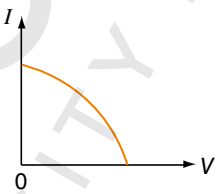
(d) The photoelectric current is always saturated when V is positive. (T/F)

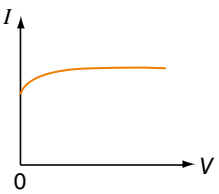
(e) When the photoelectric current becomes saturated, the number of photoelectrons emitted by the cathode is equal to the number of photoelectrons reaching the anode. (T/F)

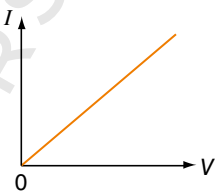
(f) When V is negative, the photoelectrons emitted from the cathode are hindered to reach the anode. (T/F)

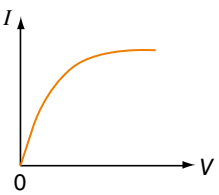
(g) No photoelectrons can reach the anode when V is negative. (T/F)

2 The value of V is adjusted and the corresponding current I is measured. What will be the graph of I against V?

A


B


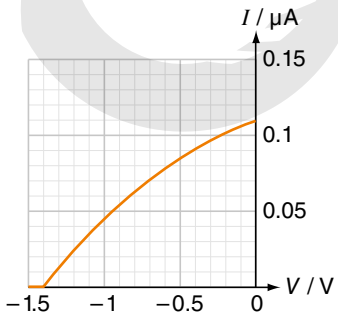
C


D


Learn by practice 1

The charge of an electron is -1.60×10^{-19} C.

1 In a photoelectric effect experiment, the variation of the photoelectric current I with the p.d. V across the photocell is as shown.



(a) Determine the stopping potential.

(b) Find the maximum KE (in eV) of the photoelectrons emitted from the cathode.

(c) What is I when $V = -0.7$ V?

(d) How many photoelectrons reach the anode in each second when $V = -0.7$ V?

2 In a photoelectric effect experiment, 8.1×10^{11} photoelectrons are emitted from the cathode in each second. What is the saturated current?

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DSE goal

State the limitations of the wave model of light in explaining the photoelectric effect.

2 Photoelectric effect and wave theory

Photoelectric effect is related to the energy transmitted by light. We use **intensity** to describe the energy transmitted by the light through each unit of area per unit time. The unit of intensity is W m^{-2}.

Power $P = \frac{E}{t}$ ▶

$$\text{Intensity} = \frac{\text{energy}}{\text{area} \times \text{time}} = \frac{E}{At} = \frac{P}{A}$$

Simulation

✎ This simulation demonstrates the various observations of the photoelectric effect. It is available on OUP web in PC version (.exe) only.

✎ While Ss show satisfactory performance in applying the photoelectric equation, they generally show poor qualitative understanding of why wave theory fails to explain the experimental observation.

Scientists studied the photoelectric effect carefully and tried to explain the phenomena they observed using the wave theory of light. Surprisingly, the observations are very different from the predictions.

a Threshold frequency

Expected result according to the wave theory ✗

According to the wave theory, an electron can accumulate the energy transmitted by the light continuously. It is emitted when it absorbs enough energy, no matter what the frequency of the light is (Fig 1.1m).

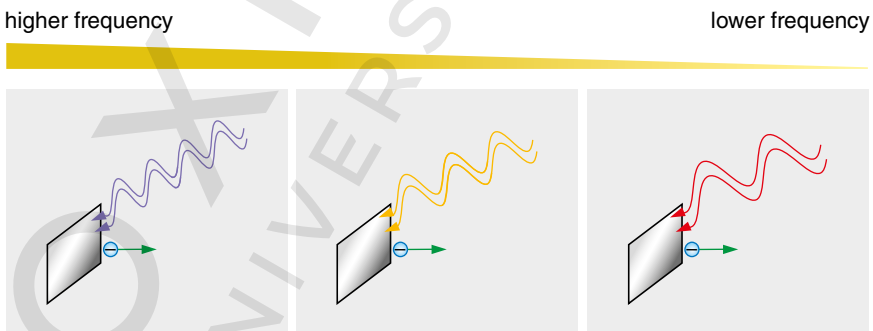


Fig 1.1m *According to the wave theory, the emission of photoelectrons is independent of the frequency of the incident light.*

Video

→ This video shows a specially treated neon bulb in a circuit lights up when it is illuminated by green light and not by orange light.

Actual observation ✓

For a given material, when the frequency of the incident light is lower than a certain minimum value, i.e. the **threshold frequency** f_0, no photoelectrons are emitted (Fig 1.1n).

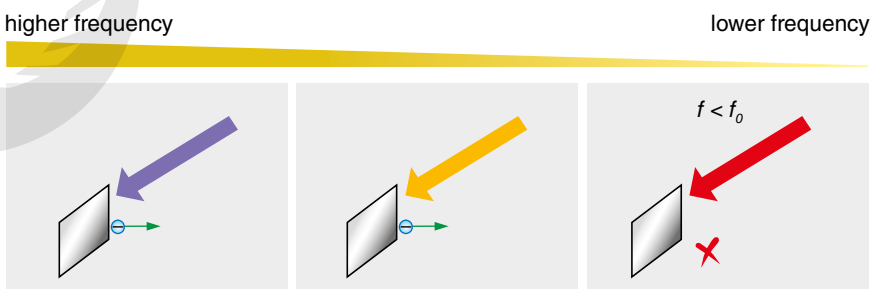


Fig 1.1n *No photoelectrons are emitted when the frequency f of the incident light is lower than the threshold frequency f_0.*

intensity 強度　　threshold frequency 臨閾頻率

Table 1.1a shows the threshold frequencies of some materials.

Material	sodium	calcium	copper	platinum
f_0 / Hz	5.6×10^{14}	7.0×10^{14}	11.3×10^{14}	15.4×10^{14}

Table 1.1a *Threshold frequencies f_0 of some materials.*

The frequency of visible light ranges from 4.3×10^{14} Hz (red) to 7.5×10^{14} Hz (violet).

The threshold frequency is the minimum frequency of light for the photoelectric effect to occur.

b No time delay

Expected result according to the wave theory ✗

According to the wave theory, the intensity of a light wave depends on the amplitude.

When the light intensity is not very high, electrons need time to accumulate enough energy from the light before leaving the surface of the material (Fig 1.1o). There should be a time delay between the arrival of light and the ejection of photoelectrons.

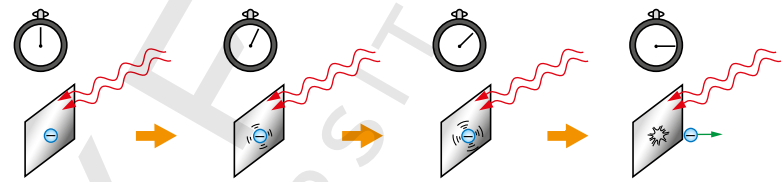


Fig 1.1o *According to the wave theory, electrons are expected to take time to leave the surface.*

Example 2 Time delay according to wave theory

A copper surface is illuminated by light of intensity 0.01 W m^{-2}. A copper atom on the surface has to absorb 7.5×10^{-19} J of energy to eject an electron. It has an effective area of 0.1 nm^2 in absorbing energy. Estimate the minimum time required for the atom to eject an electron according to the wave theory of light.

Solution

By intensity $= \frac{E}{At}$,

$$t = \frac{E}{A \times \text{intensity}} = \frac{7.5 \times 10^{-19}}{(0.1 \times 10^{-18})(0.01)} = 750 \text{ s}$$

Tip

1 nm^2 = 1 nm × 1 nm
= 10^{-9} m × 10^{-9} m
= 10^{-18} m^2

Actual observation ✓

If the frequency of light is above the threshold value, the ammeter shows a non-zero reading almost **immediately** even if the light intensity is very low. This shows that the photoelectrons are emitted with **no time delay**.

Quick check 2

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

1 A material is illuminated by light. **According to the wave theory of light**, photoelectrons will be emitted

A no matter what the frequency of the light is.

B only when the frequency of the light is higher than the threshold value.

2 When yellow light of frequency 5.1×10^{14} Hz is directed onto the surface of material X, photoelectrons are **not** emitted. Determine whether the following statements are true.

(a) If red light is directed onto the surface of X, no photoelectrons will be emitted. (T/F)

(b) If a very bright yellow light of the same frequency is directed onto the surface of X, no photoelectrons will be emitted. (T/F)

(c) The threshold frequency of X is equal to 5.1×10^{14} Hz. (T/F)

3 The threshold frequency of material P is f_0. A very dim light of frequency higher than f_0 is directed onto the surface of P. Is there any time delay between the arrival of light and the ejection of photoelectrons

(a) according to the wave theory of light,

(b) in reality?

Learn by practice 2

1 An atom on the surface of a material has to absorb 6.4×10^{-19} J of energy to eject an electron. It has an effective area of 0.1 nm^2 in absorbing energy. According to the wave theory of light, estimate the minimum time required for the atom to eject an electron when the material is illuminated by light of intensity 0.2 W m^{-2}.

c Stopping potential

Expected result according to the wave theory ✗

According to the wave theory, the rate of energy transmitted by light depends on the light intensity but not the frequency. Therefore, the maximum kinetic energy K_{max} of the photoelectrons emitted, hence the stopping potential V_s, should depend on the light intensity but not the frequency (Fig 1.1p)

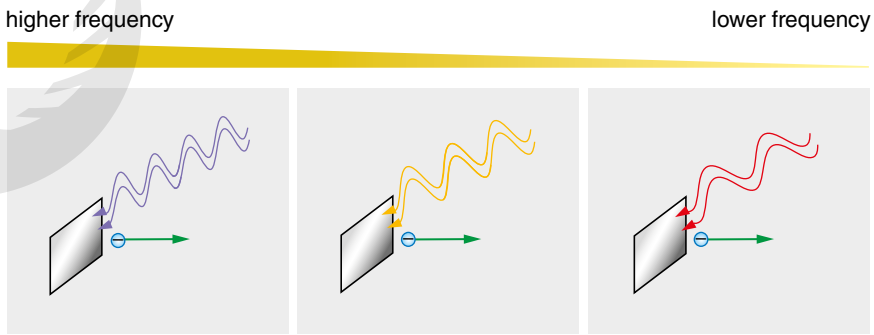


Fig 1.1p *According to the wave theory, K_{max} is expected to be independent of the frequency of the incident light.*

Know more

Philipp Lénárd (1862–1947)



Philipp Lénárd was the first scientist to discover that the energy of individual photoelectrons emitted was independent of the intensity but increased with the frequency of the incident light. He received the Nobel Prize in Physics in 1905 for his research on cathode rays (electron beams).

Actual observation ✓

For a given material, K_{max} and V_s increase with the frequency of light but are independent of the light intensity (Fig 1.1q).

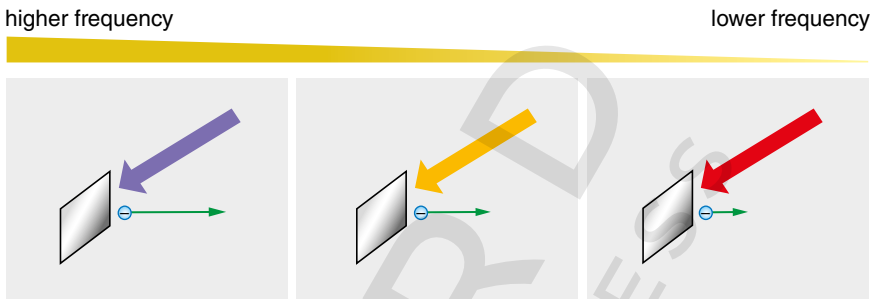


Fig 1.1q *K_{max} increases with the frequency of the incident light.*

d Saturated current

Expected result according to the wave theory ✓

The emission rate of photoelectrons, hence the saturated current, is expected to increase with the light intensity (Fig 1.1r).

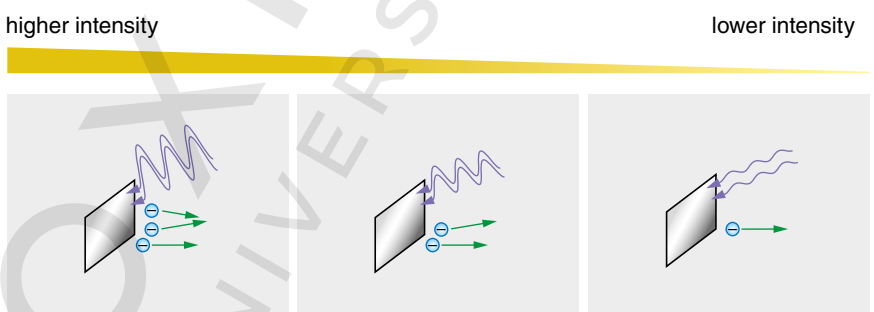


Fig 1.1r *According to the wave theory, the number of photoelectrons emitted is expected to increase with the light intensity.*

Actual observation ✓

The emission rate of photoelectrons, hence the saturated current, increases with the light intensity (Fig 1.1s).

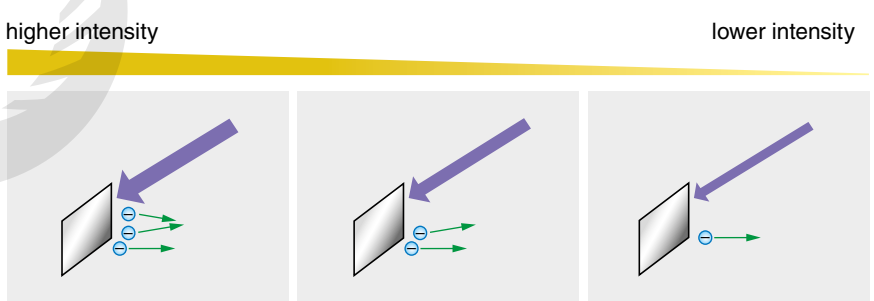


Fig 1.1s *The number of photoelectrons emitted increases with the light intensity.*

Philipp Lénárd 勒納

Figure 1.1t shows how the photoelectric current I varies with the potential difference V across the photocell for light of the same frequency but different intensities.

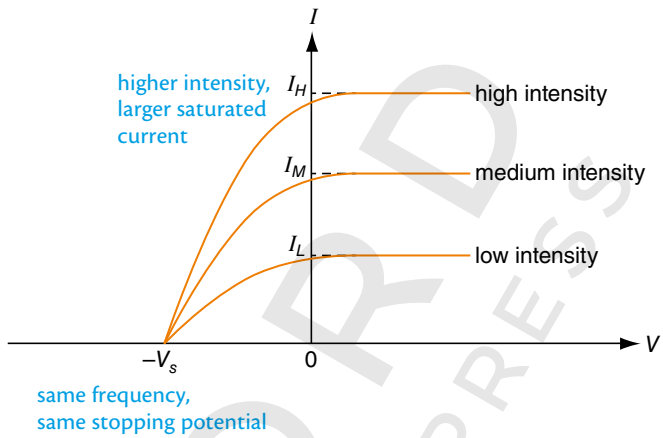


Fig 1.1t *I–V graphs for light of the same frequency but different intensities.*

We can see that some predictions of the wave theory do not match with the observations in the photoelectric effect experiment. Table 1.1b summarizes this.

	Prediction of wave theory		Observation
Frequency for emission	all frequency	✗	must above threshold frequency
Time required for emission	time is needed (with time delay)	✗	no time delay
Stopping potential	• independent of frequency • increases with intensity	✗	• increases with frequency • independent of intensity
Saturated current	increases with intensity	✓	increases with intensity

Table 1.1b *Comparing the predictions of the wave theory and the observations in the photoelectric effect experiment.*

Scientists were puzzled by these results until Einstein gave the explanations in 1905. We will learn more about this in Chapter 1.2.

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 Determine whether the following are (A) the actual results of the photoelectric experiment or (B) the prediction of the wave theory of light.

(a) The saturated current will increase if the light intensity increases.

(b) The stopping potential will increase if the light intensity increases.

(c) The stopping potential will increase if the light frequency increases.

(d) The maximum KE of the photoelectrons emitted will increase if the light intensity increases.

Exercise 1.1

(For Q1–3.) A photocell is connected as shown (Fig a). A voltage V is applied across the cathode and the anode with the cathode at a higher potential.

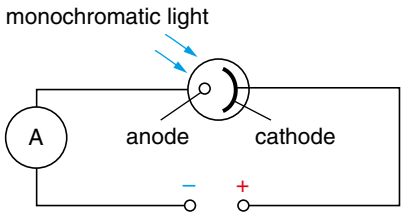


Fig a

1 If the maximum KE of the photoelectrons is 1.35 eV, which of the following sizes of V can stop all the photoelectrons from reaching the anode?

(1) 1.3 V

(2) 1.4 V

(3) 1.5 V

A (1) only B (2) only

C (1) and (2) only D (2) and (3) only

★**2** A photoelectron can reach the anode. Which of the following statements about the photoelectron must be correct?

(1) Its KE is eV when it is emitted from the cathode.

(2) Its KE decreases when it travels towards the anode.

(3) Its KE becomes zero when it reaches the anode.

A (2) only

B (3) only

C (1) and (2) only

D (2) and (3) only

★**3** If no current is detected in the circuit, which of the following must be correct?

(1) No photoelectrons are emitted from the cathode.

(2) No photoelectrons reach the anode.

(3) The frequency of the light is lower than the threshold value.

A (2) only B (3) only

C (1) and (2) only D (1), (2) and (3)

★**4** A neutral calcium plate is illuminated by light of intensity 0.02 W m^{-2} and photoelectrons are emitted. A calcium atom on the surface has to absorb 4.64×10^{-19} J of energy to eject an electron.

(a) Estimate the minimum time required for the atom to eject an electron according to the wave theory of light. Assume that a calcium atom has an effective area of 0.25 nm^2 in absorbing energy.

(b) In reality, is there any time delay for the calcium plate to emit photoelectrons?

(c) What kind of charge does the calcium plate get?

★**5** In the photoelectric effect experiment, the following graph of photoelectric current I against the p.d. V across the photocell is obtained (Fig b).

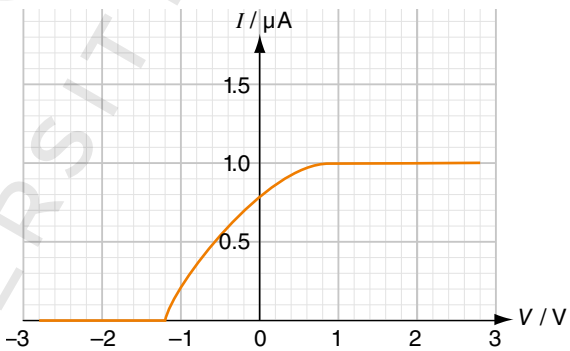


Fig b

(a) What is the stopping potential?

(b) Suggest a factor that the stopping potential depends on.

(c) What is the maximum kinetic energy (in eV) of the photoelectrons emitted?

(d) What is the emission rate of photoelectrons?

(e) On Figure b, sketch the curve if the intensity of the incident light is reduced by half while the frequency remains unchanged.

★**6** (a) State the condition required for the photoelectric effect to occur on a material surface.

(b) What is the difficulty in explaining the condition in (a) by considering light as a wave?

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1.2 Einstein's interpretation of the photoelectric effect

1 Light as particles: photons

DSE goal
State photon energy $E = hf$.

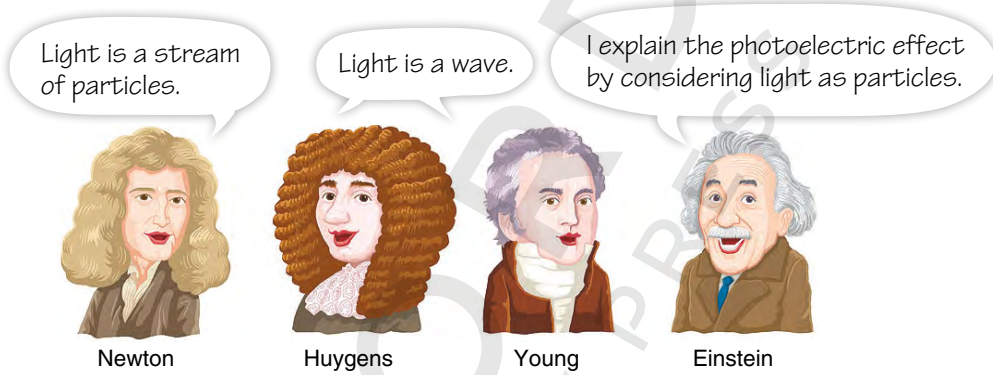


Fig 1.2a *What is the nature of light?*

Instead of thinking of light as a wave, Einstein considered light as a stream of particles, called **photons** (Fig 1.2b).



Fig 1.2b *Light as a stream of particles.*

Each photon has zero mass and carries a package, or a **quantum**, of energy which cannot be subdivided. For light of frequency f, the energy E of a photon is given by

energy of photon (J) $E = hf$ Planck constant = 6.63×10^{-34} J s; **frequency** of light (Hz)

where h is called **Planck constant** and is equal to 6.63×10^{-34} J s.

Note that the higher the frequency, the more the energy a photon carries.

Joule is too large a unit of energy to be conveniently used in the study of atomic physics. Therefore, electron-volt is usually used. Recall that

$$1 \text{ eV} = e \times 1 \text{ V} = (1.60 \times 10^{-19} \text{ C}) \times 1 \text{ V} = 1.60 \times 10^{-19} \text{ J}$$

$$1 \text{ J} = \frac{1}{1.60 \times 10^{-19}} \text{ eV}$$

photon 光子　　quantum (*pl.* quanta) 量子　　Planck constant 普朗克常數

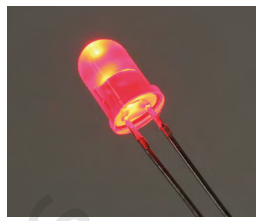
Know more

Max Planck (1858–1947)



In 1900, *Max Planck* postulated that electromagnetic energy could be emitted only in quantized form ($E = hf$). Nowadays, this assumption is regarded as the birth of *quantum physics*. Planck was awarded the Nobel Prize in Physics in 1918.

Example 3 Energy of a photon of red light



An LED emits red light. The wavelength of the red light is 700 nm in a vacuum.

Find the energy of a photon of the red light

(a) in joules,

(b) in electron-volts.

Solution

(a) $f = \frac{c}{\lambda} = \frac{3 \times 10^8}{700 \times 10^{-9}} = 4.29 \times 10^{14}$ Hz

$E = hf = (6.63 \times 10^{-34})(4.29 \times 10^{14}) = 2.84 \times 10^{-19}$ J

(b) $E = 2.84 \times 10^{-19} \times \frac{1}{1.60 \times 10^{-19}} = 1.78$ eV

Google Forms Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 Complete the table. Take $h = 6.63 \times 10^{-34}$ J s.

	f / Hz	λ / m	E / J	E / eV
Violet light	7.50×10^{14}			
Green light				2.25
Microwave		0.035		

2 *True or false:* A photon of yellow light carries more energy than a photon of red light. (T/F)

Know more

Albert Einstein and the 'Miracle Year'

In 1905, the year which is often called the 'Miracle Year', Einstein, at the age of 26, published four important papers that amounted to a significant contribution to modern physics and changed people's understanding on space, time, mass and energy. These four papers were about photoelectric effect, Brownian motion, *special relativity* and mass-energy equivalence.



Max Planck 普朗克 quantum physics 量子物理學 special relativity 狹義相對論

DSE goal

Describe how the intensity of the incident light of a given frequency is related to the number of photons.

DSE exam

14(2)Q2.6

2 Light intensity, frequency and number of photons

Consider a light beam of frequency f incident on a surface. Every photon carries the same amount of energy hf. If N photons hit the surface, the energy transmitted will be $N(hf)$. Therefore, the light intensity is directly proportional to the number of photons N hitting the surface per second (Fig 1.2c).

$$\text{Intensity} = \frac{E}{At} = \frac{N(hf)}{At} \propto \frac{N}{t}$$

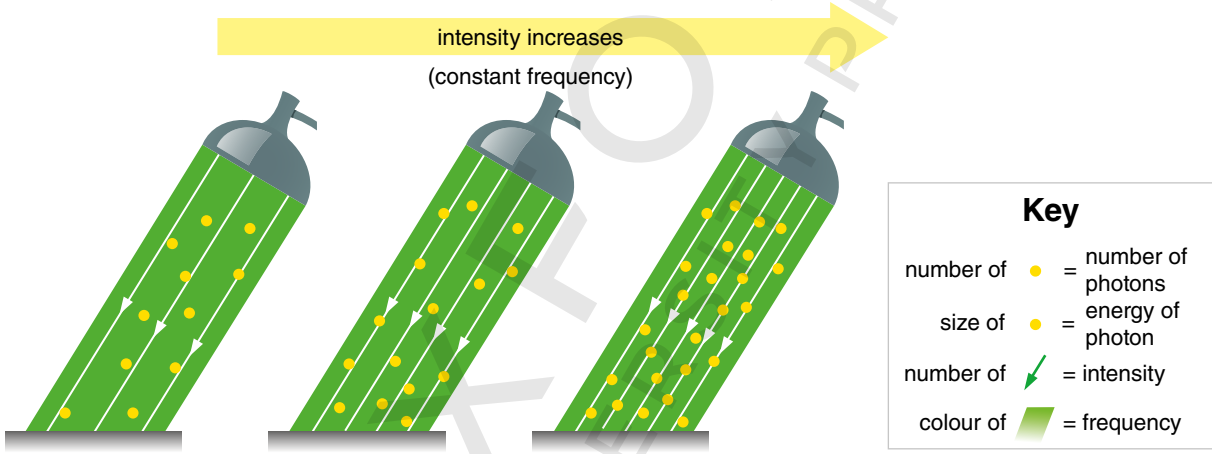


Fig 1.2c *Number of photons and light intensity.*

On the other hand, light beams of the same intensity transmit energy at the same rate. For light of a higher frequency, each photon carries more energy, so less photons are needed to transmit the same amount of energy (Fig 1.2d).

By intensity = $\frac{Nhf}{At}$, for the same intensity, $f \uparrow \Rightarrow N \downarrow$

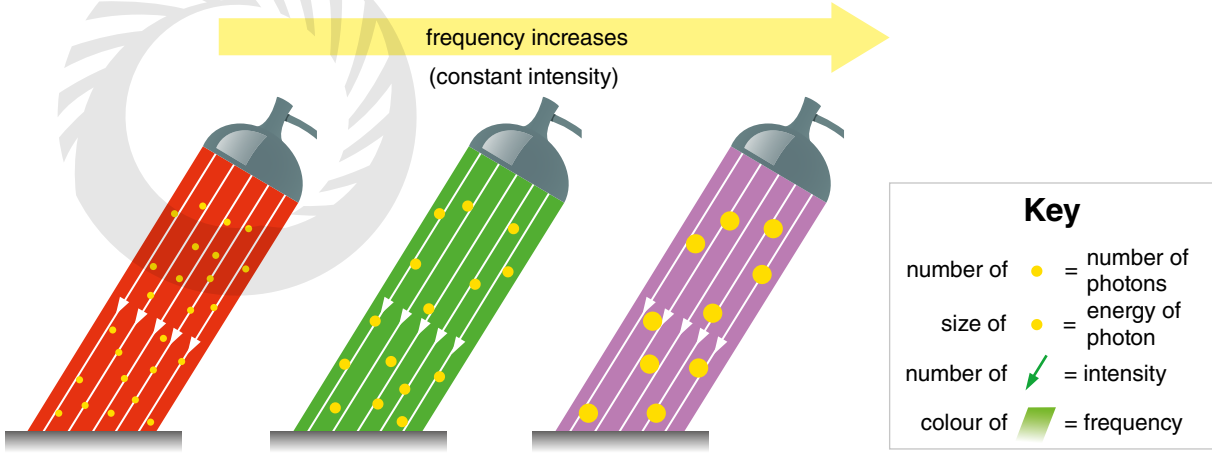


Fig 1.2d *Number of photons and frequency of light.*

Example 4 Number of photons hitting a surface in each second

Red light of frequency 4.3×10^{14} Hz and intensity 0.05 W m^{-2} falls normally on a plane surface and illuminates an area of 4×10^{-4} m^2.

(a) How many photons hit the surface per second?

(b) If the red light is replaced by blue light of frequency 6.7×10^{14} Hz and with the same intensity, how many photons hit the surface per second?

Solution

$$\text{Intensity} = \frac{E}{At} = \frac{N(hf)}{At}$$

$\frac{N}{t}$ is the number of photons hitting the surface per second. ▶

$$\therefore \frac{N}{t} = \text{intensity} \times \frac{A}{hf}$$

(a) $$\frac{N_R}{t} = \text{intensity} \times \frac{A}{hf_R}$$

$$= 0.05 \times \frac{4 \times 10^{-4}}{(6.63 \times 10^{-34})(4.3 \times 10^{14})}$$

$$= 7.02 \times 10^{13}$$

(b) $$\frac{N_B}{t} = \text{intensity} \times \frac{A}{hf_B}$$

$$= 0.05 \times \frac{4 \times 10^{-4}}{(6.63 \times 10^{-34})(6.7 \times 10^{14})}$$

$$= 4.50 \times 10^{13}$$

Google Forms Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 A light beam is incident on a surface. When the intensity of the light beam increases, the number of photons hitting the surface per second

A increases.

B decreases.

2 Two light beams of the same intensity fall normally on a plane surface and illuminate the same area. For the light beam with a higher frequency, the number of photons hitting the surface per second is

A larger.

B smaller.

3 Two light beams have the same intensity of 0.2 W m^{-2}. They fall normally on a plane surface and illuminate the same area of 5×10^{-4} m^2 in turn. Complete the table below. Take $h = 6.63 \times 10^{-34}$ J s.

	$\frac{N}{t}$	f / 10^{14} Hz
Green light	2.76×10^{14}	
Violet light		7.50

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DSE goal

Explain photoelectric effect using Einstein's photoelectric equation.

Apply $E = hf$ and Einstein's photoelectric equation to solve problems.

DSE exam

14(2)Q2.5, 16(2)Q.2, 17(2)Q2.2, Q2.3, 19(2)Q2.3, 20(2)Q.2, 21(2)Q2.3, 22(2)Q.2, etc.

3 Explaining the photoelectric effect

Einstein applied the concept of photons to explain the photoelectric effect.

a Work function

There are attractive electric forces acting between the electrons and the positively charged nuclei in a material. To make an electron escape from the material, work has to be done against the attractive force. The amount of work required depends on many factors, such as the depth of the electron from the surface of the material.

Work function (ϕ) is the minimum energy required to release an electron from the surface of a material.

Table 1.2a shows the work functions of some materials.

Material	Work function ϕ / eV
Sodium	2.3
Copper	4.7
Carbon	4.8
Platinum	6.4

Table 1.2a *Work functions of some materials.*

b One-to-one interaction

If an electron absorbs a photon, all the photon's energy will be transferred to the electron. An electron can absorb at most one photon at a time, and a photon can be absorbed by at most one electron, i.e. a one-to-one interaction. Under such an interaction, whether electrons will escape from a material's surface depends only on the energy of each incident photon and the work function of the material.

The chance for an electron absorbing two photons is so small that it can be ignored under normal conditions. ▶

c Threshold frequency

If the energy hf of the photons falling on a material is smaller than the work function ϕ of the material, no electron can escape. This implies a threshold frequency f_0 of light for the emission of photoelectrons to occur on a given material. It is related to the work function by:

$$\phi = hf_0$$

ϕ: work function of material (J); h: Planck constant = 6.63×10^{-34} J s; f_0: threshold frequency for material (Hz)

Since h is measured in J s and f_0 is measured in Hz (i.e. s^{-1}), ϕ is expressed in J when this equation is applied. ▶

work function 功函數

d No time delay

Since energy transfer takes place immediately when a photon is absorbed by an electron, no time delay is detected no matter how low the intensity of the light is.

e Saturated current

For a material receiving monochromatic light of frequency above the threshold value, when the light intensity increases, more photons will hit the material per second. More electrons in the material will absorb photons and will be emitted, so the saturated current will increase.

✎ Suppose two light beams of the same intensity fall on the cathode of a photocell in turn. The higher the frequency, the fewer the number of photons hitting the surface per second. However, these photons are more effective in producing photoelectric effect. Therefore, it is not easy to compare the saturated current due to the two light beams.

Example 5 Work function and threshold frequency of sodium

The work function of sodium is 2.3 eV.

(a) Explain whether the photoelectric effect will occur if photons, each with energy 1.78 eV, fall on a sodium plate.

(b) (i) Find the threshold frequency of sodium.

(ii) Find the longest wavelength, i.e. the threshold wavelength, λ_{max} of light that can trigger photoelectric effect on sodium.

(iii) Explain whether the photoelectric effect will occur if blue light of wavelength 450 nm is incident on a sodium plate.

(c) Why are photoelectrons emitted without time delay?

Photons with energy of 1.78 eV correspond to red light of wavelength 700 nm. See Example 3 on p.15. ▶

Solution

(a) The energy of each incident photon is lower than the work function. Therefore, the photoelectric effect will not occur.

(b) (i) By $\phi = hf_0$,

$$f_0 = \frac{\phi}{h} = \frac{2.3 \text{ eV}}{6.63 \times 10^{-34} \text{ J s}} = \frac{2.3 \times 1.60 \times 10^{-19} \text{ J}}{6.63 \times 10^{-34} \text{ J s}} = 5.551 \times 10^{14} \text{ Hz}$$

$$\approx 5.55 \times 10^{14} \text{ Hz}$$

(ii) $\lambda_{max} = \frac{c}{f_0} = \frac{3 \times 10^8}{5.551 \times 10^{14}} = 5.40 \times 10^{-7} \text{ m}$

(iii) Since the wavelength of the blue light is shorter than λ_{max}, the photoelectric effect will occur.

(c) An electron can be emitted immediately if it absorbs a photon with enough energy.

Quick check 6

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

1 *True or false:*

(a) If the photons falling on a material have energy lower than the work function, no photoelectron will be emitted. (T/F)

(b) The energy of a photon can be shared between two electrons. (T/F)

(c) Photoelectrons are emitted with no time delay because when a photon is absorbed by an electron, all energy of the photon is transferred to the electron immediately. (T/F)

(d) The energy required by all electrons to leave a material is equal to the work function. (T/F)

(e) It is common for an electron to absorb more than one photon before escaping from a material. (T/F)

2 Complete the table. Given: $h = 6.63 \times 10^{-34}$ J s

Metal	ϕ / eV	f_0 / 10^{14} Hz
Calcium	2.9	
Gold		12.3

Learn by practice 3

Given: $h = 6.63 \times 10^{-34}$ J s

1 Find the maximum wavelength of light that can trigger the photoelectric effect on a material of work function 3.1 eV.

2 Light of wavelength 500 nm falls on a material of work function 3.5 eV. Will the photoelectric effect occur?

f Einstein's photoelectric equation

After an electron absorbs a photon with high enough energy hf, it escapes from the material by using up some of the energy. The remaining energy, if any, will become the kinetic energy of the electron, i.e.

$$\text{Energy of photon} = \begin{matrix}\text{energy required for electron} \\ \text{to escape from material}\end{matrix} + \text{KE of electron}$$

The electrons emitted have different kinetic energy as they require different amounts of energy to escape. The electrons which require the least amount of energy to escape (equal to the work function ϕ) will gain the maximum kinetic energy K_{max}. Therefore,

$$hf = \phi + K_{max}$$

$$K_{max} = hf - \phi = h(f - f_0)$$

- K_{max}: max. KE of photoelectrons (J)
- f: frequency of light (Hz)
- ϕ: work function of material (J)
- f_0: threshold frequency of material (Hz)
- h: Planck constant = 6.63×10^{-34} J s

$\phi = hf_0$ ▶

Also, by $c = f\lambda$,

$K_{max} = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_{max}}\right)$

This is **Einstein's photoelectric equation**.

Einstein's photoelectric equation 愛因斯坦光電方程

The equation can be extended to describe the maximum speed v_{max} of the photoelectrons (each with mass m_e) emitted:

$$\frac{1}{2}m_e v_{max}^2 = hf - \phi$$

From Einstein's photoelectric equation, it is clear that K_{max} increases with the frequency f of the incident light (Fig 1.2e).

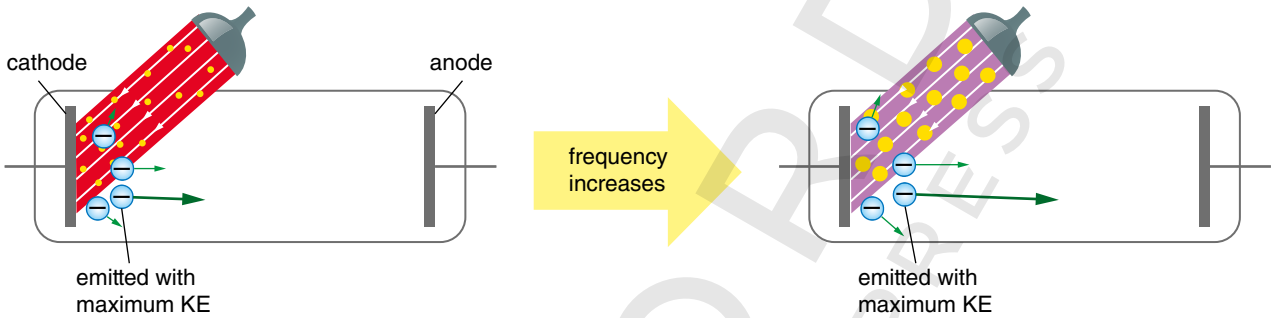


Fig 1.2e *Effect of changing the frequency of light.*

Increasing the light intensity while keeping the frequency unchanged will only increase the number of photons reaching the material surface, but not the energy of each photon, so K_{max} will remain the same.

Example 6 | Maximum KE of photoelectrons

The work function of a metal is 2.49 eV. Light of frequency 7.5×10^{14} Hz is incident on the metal. The mass of an electron is 9.11×10^{-31} kg.

(a) Find the maximum kinetic energy K_{max} (in J) of the photoelectrons.

(b) Find the highest speed v_{max} of the photoelectrons emitted.

(c) How will K_{max} change if light of a higher frequency is used?

(d) Why are photoelectrons emitted with different kinetic energy but not only the value in (a)?

Solution

Tip
Express ϕ in J.

(a) $K_{max} = hf - \phi$

$= (6.63 \times 10^{-34})\ (7.5 \times 10^{14}) - (2.49)\ (1.60 \times 10^{-19})$

$= 9.89 \times 10^{-20}$ J

(b) By $K_{max} = \frac{1}{2}m_e v_{max}^2$,

$$v_{max} = \sqrt{\frac{2K_{max}}{m_e}} = \sqrt{\frac{2\ (9.89 \times 10^{-20})}{9.11 \times 10^{-31}}} = 4.66 \times 10^5 \text{ m s}^{-1}$$

Tip
By $K_{max} = hf - \phi$, $f \uparrow \Rightarrow K_{max} \uparrow$

(c) Increase

(d) Electrons not at the surface need energy more than the work function to escape.

Google Forms Microsoft Forms

Quick check 7

→ E-version in OUP Exercise Platform is also available.

1 Complete the table.

	K_{max} / eV	ϕ / eV	hf / eV
(a)	2	3	
(b)		2	6

2 Complete the table. Take $h = 6.63 \times 10^{-34}$ J s.

	K_{max} / 10^{-19} J	f_0 / 10^{14} Hz	f / 10^{14} Hz
(a)	2	7	
(b)		5	6

Learn by practice 4

Take $h = 6.63 \times 10^{-34}$ J s, $m_e = 9.11 \times 10^{-31}$ kg.

1 (a) What is the meaning of the work function of a material?

(b) When a material is illuminated by light of wavelength 3×10^{-7} m, the maximum kinetic energy of the photoelectrons emitted is 0.8 eV. Find the work function of the material.

2 When light of frequency f falls on a material of work function 2 eV, the maximum kinetic energy of the photoelectrons is 3 eV. Find f.

3 What is the maximum kinetic energy (in eV) of the photoelectrons if photons of energy 6 eV hit a copper plate? The work function of copper is 4.7 eV.

4 Violet light of frequency 7.5×10^{14} Hz is directed onto a calcium plate. The threshold frequency of calcium is 7×10^{14} Hz.

(a) Find the highest speed of the photoelectrons emitted.

(b) Why are the photoelectrons emitted with different KE?

Table 1.2b summarizes how Einstein's photon model successfully explains the observations in the photoelectric effect experiment.

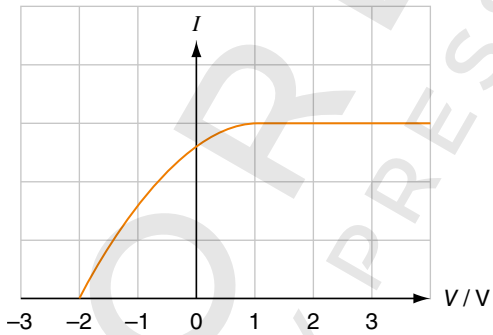
Observation	Explanation by photon model
There is a threshold frequency below which no photoelectron is emitted.	The energy hf of a photon must be higher than the work function of the material for an electron to be emitted.
Photoelectrons are emitted without time delay.	An electron is emitted once it absorbs a photon with high enough energy.
V_s and K_{max} increase with the frequency of the light but are independent of the light intensity.	$eV_s = K_{max} = hf - \phi$ Increasing the intensity only increases the number of incident photons, not the energy of the photons.
The emission rate of photoelectrons, hence the saturated current, increases with the light intensity.	The number of incident photons increases with the light intensity.

Table 1.2b *Observations explained by the photon model.*

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Example 7 | *I–V* graph

In a photoelectric effect experiment, a monochromatic light beam is directed onto the cathode of a photocell. Each photon of the light carries energy of 4.5 eV. The photoelectric current I varies with the potential difference V across the photocell as shown. V is taken as positive when the anode of the photocell is connected to the positive terminal of the power supply.



(a) Find the work function, in eV, of the cathode of the photocell.

(b) If the light intensity increases but the frequency remains unchanged, sketch the new curve obtained.

(c) If the experiment is repeated with light of frequency 7×10^{14} Hz, what will be the stopping potential?

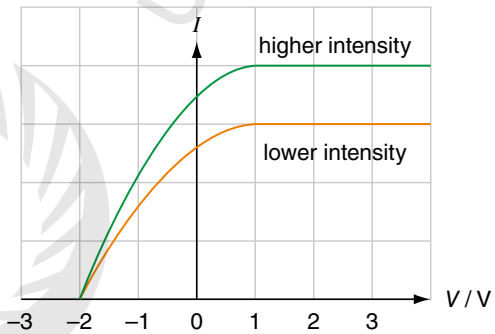
Solution

(a) $K_{max} = eV_s = e \times 2 \text{ V} = 2 \text{ eV}$

By $K_{max} = hf - \phi$,

$\phi = hf - K_{max} = 4.5 - 2 = 2.5 \text{ eV}$

(b)



Tip
Express hf in eV.

(c) $K_{max} = hf - \phi = (6.63 \times 10^{-34})(7 \times 10^{14})\left(\frac{1}{1.60 \times 10^{-19}}\right) - 2.5 = 0.401 \text{ eV}$

$V_s = \frac{K_{max}}{e} = \frac{0.401 \text{ eV}}{e} = 0.401 \text{ V}$

▶ Exercise 1.2 Q10 (p.31)

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☆ HKDSE 2016 Paper 2 Q.2, 2017 Paper 2 Q2.2

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Work function and maximum kinetic energy

In a photoelectric effect experiment (Fig a), the cathode of the photocell is illuminated by monochromatic light of frequency f. The p.d. V across the photocell is varied and the photoelectric current I is measured. The I–V graph obtained is as shown (Fig b).

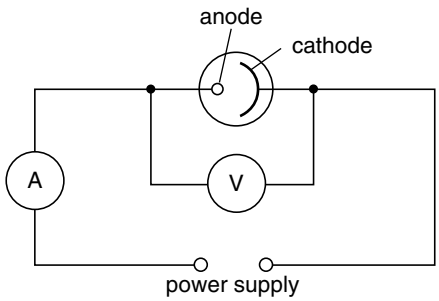


Fig a

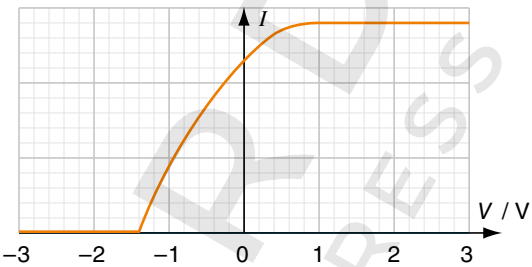


Fig b

Which of the following statements must be correct?

(1) The work function of the cathode is 1.4 eV.

(2) When I is just saturated, the maximum kinetic energy of the photoelectrons emitted from the cathode is 1.4 eV.

(3) When I is just saturated, the maximum kinetic energy of the photoelectrons reaching the anode is 1.4 eV.

A (1) only B (2) only

C (2) and (3) only D (1), (2) and (3)

Solution

From the graph, the stopping potential V_s = 1.4 V

$K_{max} = eV_s = e \times 1.4\text{ V} = 1.4\text{ eV}$

K_{max} depends on ϕ and f but not V.

∴ (2) is correct.

I is just saturated when V = +1.0 V. At this p.d., a photoelectron gains KE of 1.0 eV when it moves from the cathode to the anode (Fig c).
Therefore, maximum KE reaching anode = 1.4 + 1.0 = 2.4 eV

Common mistake
Students may not understand the meaning of the max. KE of photoelectrons reaching the anode and how it is affected by V. Some may wrongly think that it is equal to K_{max}. Some may make mistakes in the sign of V.

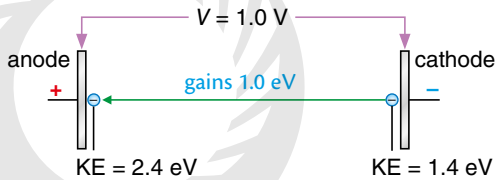


Fig c

∴ (3) is incorrect.

The work function ϕ of the cathode can be found by $K_{max} = hf - \phi$. As the value of hf, i.e. the energy of the incident photon, is not known, the value of ϕ cannot be calculated.

Common mistake
Some students may fail to differentiate K_{max}, ϕ and V_s.

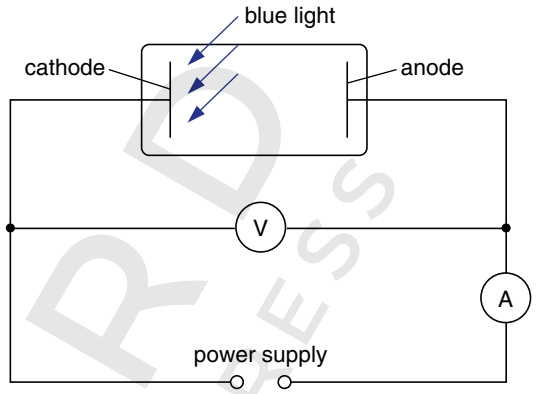
∴ (1) may not be correct.

∴ The answer is B.

▶ Revision 1 Q13 (p.36)

Writing practice

In a photoelectric effect experiment, when the photocell is illuminated by blue light, a photoelectric current is measured.



☆ HKDSE 2020 Paper 2 Q.2(b)

Question 1

The output voltage of the power supply is adjusted until the ammeter reading drops to zero. How will the ammeter reading change if the intensity of the blue light increases? Explain. (2 marks)

Answer

The ammeter reading remains zero. 1A

Increasing the intensity will not increase the energy carried by each incident photon. The maximum KE of photoelectrons emitted will not change and the photoelectrons still cannot reach the anode. 1A

Question 2

How will the stopping potential change if ultra-violet radiation is used instead of blue light? Explain. (3 marks)

Answer

The stopping potential will be higher. 1A

A photon of ultra-violet radiation has higher energy than a photon of blue light. 1A

The maximum KE of photoelectrons emitted will be higher, resulting in a higher stopping potential. 1A

Practise yourself

In a photoelectric effect experiment, when the photocell is illuminated by blue light, a photoelectric current is measured.

1 How will the stopping potential change if the intensity of the blue light increases? Explain. (2 marks)

2 Photoelectric current is measured when green light is used instead of blue light. How does the stopping potential change when green light is used? Explain. (3 marks)

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DSE exam
12(2)Q2.4, 15(2)Q2.3, 18(2)Q.2, 21(2)Q2.4

4 Verification of Einstein's model

In 1916, *Robert Millikan* performed experiments to verify whether Einstein's model of the photoelectric effect is correct.

In these experiments, monochromatic lights of different frequencies f illuminated the same metal in turn and the corresponding stopping potentials V_s were measured. When V_s was plotted against f, a straight line which did not pass through the origin was obtained (Fig 1.2f).



Fig 1.2f *Stopping potential V_s versus frequency f of incident light.*

Recall that $K_{max} = eV_s$ and $K_{max} = hf - \phi$,

$$eV_s = hf - \phi$$

Planck constant = 6.63×10^{-34} J s

work function of material (J)

stopping potential (V) $\rightarrow V_s = \left(\frac{h}{e}\right)f - \left(\frac{\phi}{e}\right)$

magnitude of charge of electron = 1.60×10^{-19} C

frequency of light (Hz)

From this equation, if Einstein's model works, the slope of the graph in Figure 1.2f should be equal to h/e. The value of Planck constant h can then be obtained by multiplying the slope by e.

The value of h obtained by Millikan agreed with the presently accepted value (differing by about 0.5% only). This result supported Einstein's model of the photoelectric effect.

When different metals are used in the experiment, straight lines with the **same slope** (equal to h/e) but different x-intercepts would be obtained (Fig 1.2g on p.27). The ***x*-intercepts** are equal to the **threshold frequencies** f_0 for the metals. According to work function $\phi = hf_0$, the larger the value of ϕ, the greater the x-intercept (higher f_0).

When $V_s = 0$,

$$0 = \left(\frac{h}{e}\right)f - \left(\frac{\phi}{e}\right)$$

$$f = \frac{\phi}{h} = f_0$$

Note that when f is smaller than f_0, no photoelectron is emitted. Therefore, the graph can only be drawn for f larger than or equal to f_0.

Know more

Robert Millikan (1868–1953)



Millikan originally set out to prove that Einstein was wrong. Ironically, his experiments confirmed Einstein's model. He received the Nobel Prize in Physics in 1923 for successfully measuring the charge of an electron.

Robert Millikan 密立根

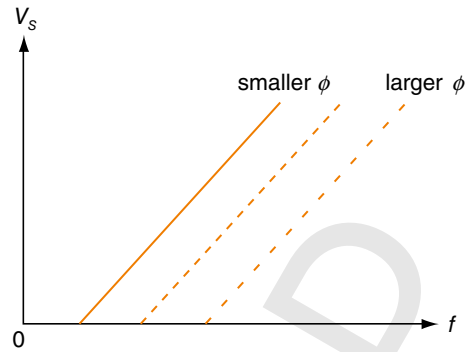
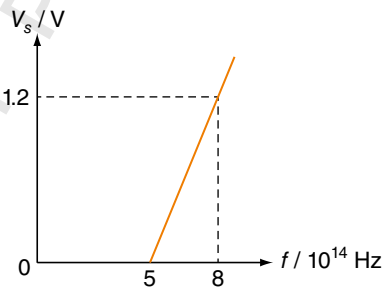


Fig 1.2g *The slopes for different metals are the same while the x-intercepts depend on the work functions.*

Example 8 | V_s–f graph

In a photoelectric effect experiment, lights of different frequencies f are incident on metal A in turn and the corresponding stopping potentials V_s are recorded. The V_s–f graph obtained is as shown.



(a) Estimate the threshold frequency for metal A.

(b) Find the experimental value of Planck constant from the graph.

(c) Estimate the work function (in eV) of metal A using the value obtained in (b).

(d) The experiment is repeated with metal B with a smaller work function. Sketch the V_s–f graph for metal B.

Solution

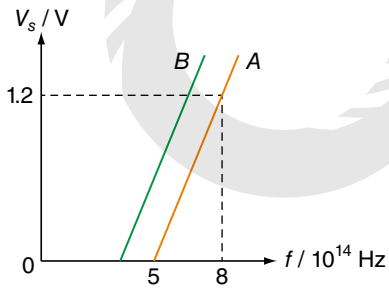
(a) 5×10^{14} Hz

(b) Slope of graph $= \dfrac{h}{e}$

$$h = \text{slope} \times e = \frac{1.2 - 0}{(8 - 5) \times 10^{14}} \times 1.60 \times 10^{-19} = 6.4 \times 10^{-34} \text{ J s}$$

(c) $\phi = hf_0 = \dfrac{(6.4 \times 10^{-34})(5 \times 10^{14})}{1.60 \times 10^{-19}} = 2 \text{ eV}$

(d)



Caution

Do not use the point (0, 0) in finding the slope. The line does not pass through the origin.

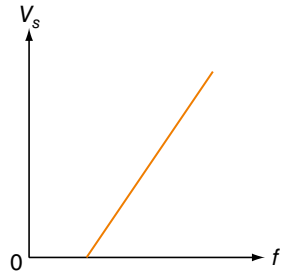
Pay attention to the factor 10^{14} for f.

Quick check 8

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

1 In a photoelectric effect experiment, lights of different frequencies f are incident on the same metal in turn. A graph of the stopping potential V_s against f is obtained.



(a) What is the slope of the graph?

A f_0 B h

C $\frac{h}{e}$ D $\frac{\phi}{e}$

(b) What is the x-intercept of the graph?

A f_0 B h

C $\frac{h}{e}$ D $\frac{\phi}{e}$

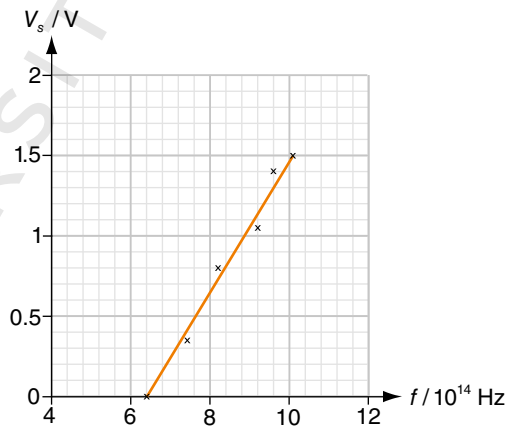
Learn by practice 5

1 In a photoelectric effect experiment, lights of different frequencies f are incident on the same metal in turn. A graph of the stopping potential V_s against f is plotted.

(a) Find the threshold frequency for this metal.

(b) Estimate Planck constant from the graph.

(c) The experiment is repeated with another metal with a larger work function. Sketch the new V_s–f graph.



DSE goal

Realise the photoelectric effect as the evidence of particle nature of light.

5 Wave-particle duality of light

The photoelectric effect gives strong support for the particle nature of light. However, the wave nature of light is also strongly supported by experiments (e.g. Young's double slit experiment). Scientists are forced to accept that light exhibits both wave and particle natures, and call this the **wave-particle duality** of light. This dual nature is not limited to visible light, but applies to all kinds of electromagnetic radiation.

Light exhibits wave-particle duality.

wave-particle duality 波粒二象性

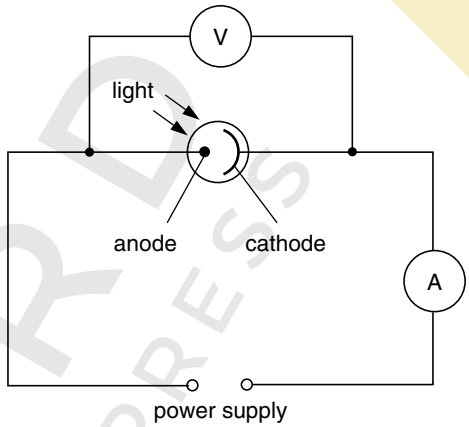
☆ HKDSE 2018 Paper 2 Q.2

→ This video demonstrates how this question is solved.

Video

DSE practice 2 | K_{max} and λ

In the photoelectric effect experiment, lights of different wavelengths λ are directed onto the cathode of a photocell in turn.



(a) How can the maximum kinetic energy K_{max} of the photoelectrons emitted from the cathode be measured using this set-up for light with a certain wavelength? (2 marks)

(b) Sketch a straight-line graph relating K_{max} and λ. (2 marks)

(c) How can you estimate Planck constant h and the threshold frequency f_0 from the graph in (b)? (2 marks)

(d) How will the graph in (b) be affected if the experiment is repeated with light of higher intensity? Explain. (2 marks)

(e) Photoelectrons are emitted without time delay even if the intensity of the light is low. State the implication of this result. (1 mark)

Solution

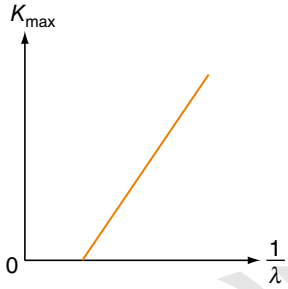
(a) Adjust the output voltage of the power supply until the ammeter reading just drops to zero. 1A

Common mistake
Students may just state the condition 'the ammeter reading just drops to zero' without describing how to achieve this.

The reading of the voltmeter gives the stopping potential V_s. K_{max} is equal to eV_s. 1A

Common mistake
Students may wrongly take V_s as the max KE.

(b)



$K_{max} = hf - \phi = \frac{hc}{\lambda} - \phi$

(Correct axes) 1A

(Correct straight line not through origin) 1A

(c) Planck constant h can be found by dividing the slope by the speed of light c. 1A

Slope $= hc$

$h = \frac{\text{slope}}{c}$

The threshold frequency f_0 can be found by multiplying the x-intercept by c. 1A

$x\text{-intercept} = \frac{\phi}{hc} = \frac{f_0}{c}$

$f_0 = (x\text{-intercept}) \times c$

(d) The graph will be the same 1A

because K_{max} is not affected by the intensity of light. 1A

Common mistake
Students may not be able to clearly state the reason (K_{max} depends only on f but not the intensity).

(e) Energy of light is transmitted in quanta. 1A

▶ Revision 1 Q25 (p.39)

Exercise 1.2

Take $c = 3.00 \times 10^8$ m s^{-1}, $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg and $h = 6.63 \times 10^{-34}$ J s.

★**1** Which of the following statements about the work function ϕ and the threshold frequency f_0 of a metal is/are correct?

(1) The frequency of the incident light should be higher than f_0 for the photoelectric effect to occur.

(2) The intensity of the incident light should be higher than ϕ for the photoelectric effect to occur.

(3) A metal with a higher value of ϕ has a higher value of f_0.

A (1) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★**2** The work functions of four metals are shown in Table a.

Metal	Work function (eV)
gold	5.1
iron	4.5
uranium	3.6
caesium	2.1

Table a

Monochromatic light of wavelength 486 nm is incident on these metals. How many of them would emit photoelectrons?

A 0 B 1

C 2 D 3

★**3** Light is incident on a metal surface. What will happen if the wavelength of light increases but the intensity remains unchanged?

(1) The number of photons hitting the surface per second increases.

(2) The energy transmitted to the surface per second increases.

(3) The average speed of the photoelectrons emitted from the metal surface increases.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

★**4** In the photoelectric effect experiment, the anode is connected to the positive terminal of the power supply (Fig a). A monochromatic light is directed onto the cathode and a photoelectric current is produced.

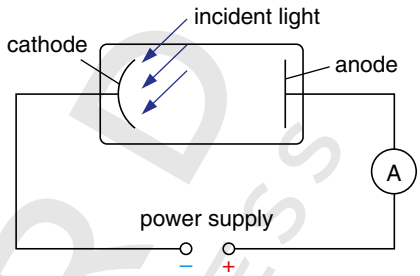


Fig a

Which of the following will occur if the output voltage of the power supply decreases in magnitude?

A The photoelectric current increases.

B The number of photoelectrons emitted from the cathode increases.

C The maximum KE of the photoelectrons reaching the anode decreases.

D The maximum KE of the photoelectrons emitted from the cathode increases.

★**5** Lasers are commonly used in light shows, e.g. A Symphony of Lights (Fig b). If a green laser beam with wavelength 532 nm is emitted at a rate of 7 W, what is the number of photons emitted per second?



Fig b

A 1.3×10^7 B 3.9×10^{15}

C 1.9×10^{19} D 2.0×10^{40}

★**6** Light is directed onto a metal plate so that the photoelectric effect occurs.

(a) Is there any time delay between the arrival of light and the emission of photoelectrons?

(b) How did Einstein explain the result in (a)?

(c) The maximum KE of the photoelectrons emitted is independent of the intensity of the light. How did Einstein explain this?

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★**7** Figure c shows the set-up used to study the photoelectric effect. When red light is shone onto the cathode, no current is detected. John and Tom are discussing the result.

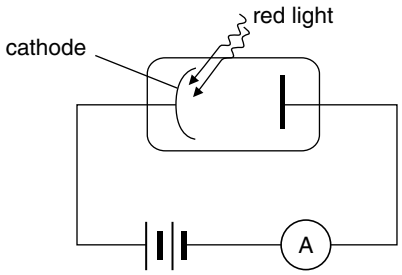


Fig c

(a) John says, 'It is possible that no photoelectrons are emitted from the cathode.'

(i) What is the necessary condition for photoelectrons to be emitted when light is directed onto the cathode?

(ii) Hence suggest a change to the set-up which can produce a photoelectric current for the situation told by John.

(b) Tom says, 'It is possible that photoelectrons are emitted but no current is produced.'

(i) Explain briefly why Tom thinks this is so.

(ii) Hence suggest a change to the set-up which can produce a photoelectric current for the situation told by Tom.

★**8** The graph below shows the results of a photoelectric effect experiment for metal Y (Fig d).

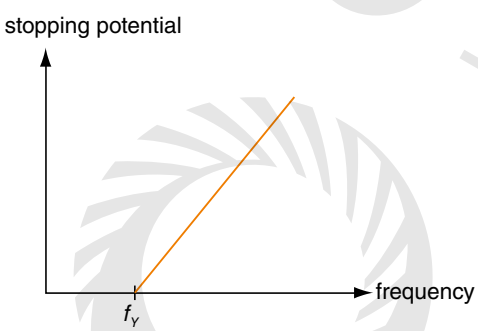


Fig d

(a) What does f_Y represent?

(b) How did Einstein explain the existence of the physical quantity in (a)?

(c) Metal Z, with a larger work function than Y, is then used in the experiment. Sketch the result in Figure d.

★**9** Figure e shows the results of the photoelectric effect experiment. K_{max} is the maximum kinetic energy of the photoelectrons emitted from the cathode when light of frequency f is incident on the cathode.

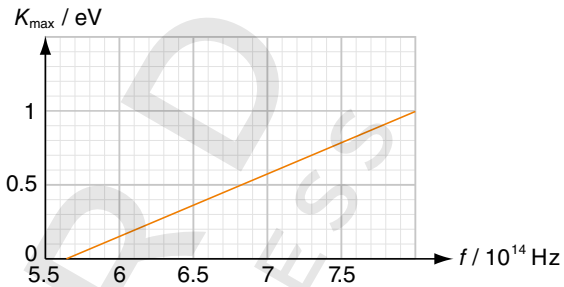


Fig e

(a) Estimate the value of Planck constant h from the graph.

(b) Estimate the work function of the cathode using the answer to (a).

★**10** In a photoelectric effect experiment, when the cathode of the photocell is illuminated by a light beam, photoelectric current I flows through the circuit. The energy of each photon of the light is 3 eV. Figure f shows how I varies with the voltage V across the photocell.

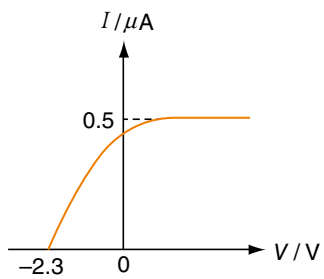


Fig f

(a) Find the wavelength of the light. What kind of electromagnetic radiation is it?

(b) Find the maximum speed of the photoelectrons emitted from the cathode.

(c) Find the work function of the cathode.

(d) How many photoelectrons are emitted from the cathode in each second?

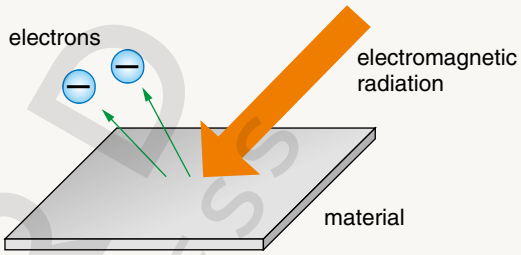
(e) Suggest two factors that affect the maximum kinetic energy of the photoelectrons reaching the anode.

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Key concepts 1

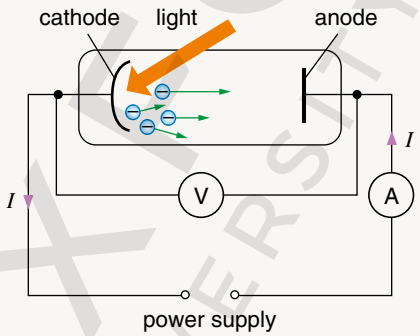
1 Photoelectric effect

- emission of electrons (photoelectrons) from the surface of a material exposed to EM radiation
- a **threshold** (minimum) **frequency** f_0 for the photoelectric effect to occur for a given material
- no time delay
- intensity of light ↑ ⇒ rate of emission ↑
- frequency of light ↑ ⇒ max. KE of photoelectrons ↑



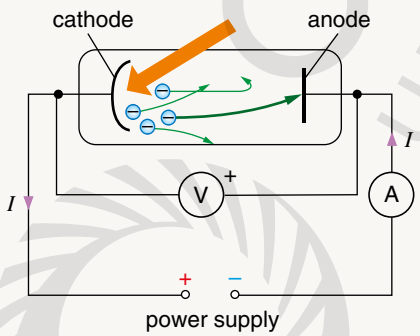
2 Photoelectric effect experiment

- Photoelectric current is measured if photoelectrons reach the anode

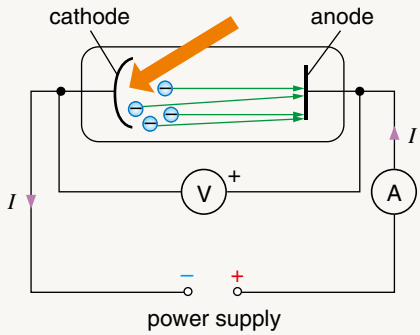


Q24 Exam report: In (a)(i), many candidates were not aware that the current saturates at a voltage of 1.0 V (which is larger than 0 V) for the photocell concerned. Candidates did not fully understand what (a)(ii) asked for and very few obtained the correct answer using the information from the graph. In answering (b)(i), some mistook the result found in (a)(ii) as the work function and a lot of them failed to differentiate maximum kinetic energy, work function and stopping potential. Candidates were able to apply $\lambda = \frac{hc}{\phi}$ to find λ but some forgot to convert the unit eV to J or just found the threshold frequency instead of wavelength. In (b)(ii), most candidates knew that the larger the wavelength, the smaller the energy of the source and thus, were able to deduce a correct conclusion. Part (c) was well answered though some candidates thought that a greater saturation current is the result of increased frequency.

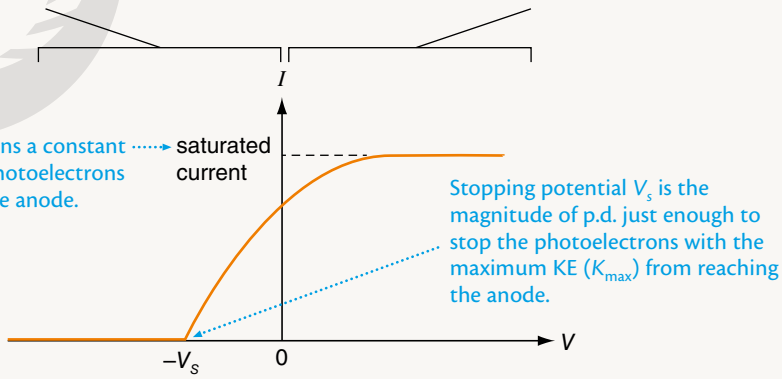
- Graph of photoelectric current I against p.d. V:



The p.d. is against the motion of photoelectrons.



The p.d. promotes photoelectrons to reach the anode.



The current attains a constant value when all photoelectrons emitted reach the anode.

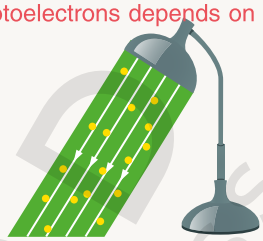
Stopping potential V_s is the magnitude of p.d. just enough to stop the photoelectrons with the maximum KE (K_{max}) from reaching the anode.

Q25 Exam report: Candidates were in general weak at presenting qualitative answers like those required in (a). Many just stated 'no current passing through the circuit' without describing the procedures to achieve that condition. Some thought that the voltage V_s was the maximum kinetic energy of the photoelectrons. In (b)(i), most candidates were able to find the wavelength correctly but some could not tell the radiation was ultra-violet. In (b)(ii), most candidates found the slope correctly but a lot of them gave incorrect units or forgot to include 10^{14} for the frequency in their calculation. Weaker ones just chose one point from the graph together with (0, 0) to find the slope. Not many knew the physical meaning of the slope and some wrongly stated the slope as h/e. In (b)(iii), most candidates knew how to find the work function. However, some of them gave the answer in J or wrongly read the threshold frequency from the graph. In (c), most candidates knew that the graph remained unchanged but not many were able to correctly explain the concept 'maximum kinetic energy of photoelectrons depends on frequency only' of photoelectric effect.

3 Particle nature of light

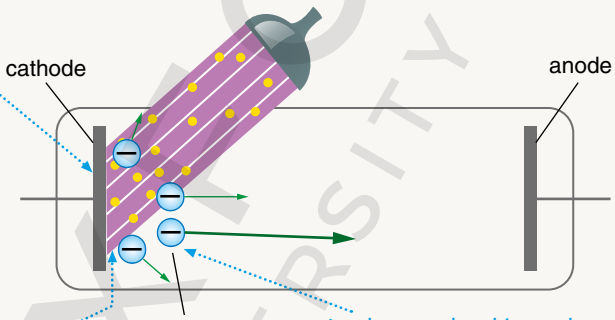
Photons

- quanta of light
- each carries energy $E = hf$
- intensity of light $\propto$ number of photons



4 Explanation of photoelectric effect

Each material has its own **work function** ϕ, i.e. the minimum energy required to release an electron from it.

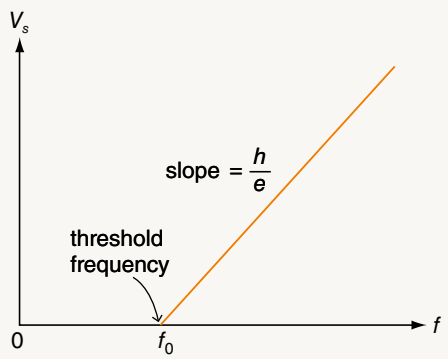


A photon may release a photoelectron if its energy is larger than ϕ. Threshold frequency f_0 is given by $\phi = hf_0$

An electron absorbing a photon uses up some of its energy to escape. The remaining energy becomes its KE. Among all photoelectrons, $K_{max} = hf - \phi = h(f - f_0)$ This is **Einstein's photoelectric equation.**

5 Graph of V_s against f

$$\begin{cases} K_{max} = eV_s \\ K_{max} = hf - \phi \end{cases} \Rightarrow V_s = \left(\frac{h}{e}\right)f - \left(\frac{\phi}{e}\right)$$



Q26 Exam report: Part (a)(i) was well answered. Many candidates misunderstood (a)(ii) and tried to explain why there was such an experimental result instead of giving the implication of the result. In (b)(i), many gave answers like 'current increases due to more photoelectrons' for the situation prior to the voltage being set to 1.7 V but without mentioning what would happen when V = 1.7 V. Candidates did well in (b)(ii). However, some simply took the photon energy hf as the work function or failed to deal the unit eV. In (c)(i), some candidates did not realise that the number of photoelectrons could simply be found from the current. In (c)(ii), not many fully understood how 'stopping potential' and 'maximum kinetic energy of photoelectrons' were related. As a result, few were able to work out the maximum kinetic energy. It seemed that candidates did not have a clear picture of the photoemission process involving the concept of work function. Some thought that the variation in kinetic energy was due to things that had happened when the photoelectrons went from cathode to anode.

Key terms

1 Einstein's photoelectric equation 愛因斯坦光電方程 p.20
2 intensity 強度 p.8
3 photocell 光電池 p.3
4 photoelectric current 光電流 p.4
5 photoelectric effect 光電效應 p.2
6 photoelectron 光電子 p.2
7 photon 光子 p.14
8 Planck constant 普朗克常數 p.14
9 quantum (pl. quanta) 量子 p.14
10 saturated current 飽和電流 p.4
11 stopping potential 遏止電勢 p.5
12 threshold frequency 臨閾頻率 p.8
13 wave-particle duality 波粒二象性 p.28
14 work function 功函數 p.18

Revision 1

Take $c = 3.00 \times 10^8$ m s^{-1}, $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg and $h = 6.63 \times 10^{-34}$ J s.

Multiple-choice questions

★1 1.2 In the photoelectric effect experiment, different p.d. are applied across the cathode and the anode in the photocell. The results are recorded in Table a.

P.d. / V	1	1.5	2	2.5
Is current detected?	✓	✓	✗	✗

Table a

Which of the following is a possible value of the maximum kinetic energy of the photoelectrons emitted?

A 1.7×10^{-19} J B 2.3×10^{-19} J
C 2.6×10^{-19} J D 3.3×10^{-19} J

★2 1.2 An electron absorbs a photon and escapes from a metal of work function ϕ. Which of the following statements must be correct?

(1) The energy required by the electron to escape from the metal is ϕ.
(2) The electron gains an energy ϕ after absorbing the photon.
(3) All the energy of the photon is transferred to the electron.

A (2) only B (3) only
C (1) and (3) only D (1), (2) and (3)

★3 1.2 When a photon travels from air into water, its energy

A increases.
B decreases.
C remains unchanged.
D changes depending on the angle of incidence.

★4 1.1 In the photoelectric effect experiment, light of frequency f is directed onto the cathode in the photocell. The p.d. V between the cathode and the anode is varied and the photoelectric current I is measured. The graph below shows how I changes with V (Fig a).

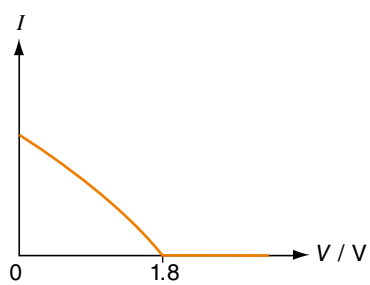


Fig a

Which of the following statements is/are correct?

(1) The threshold frequency is higher than f.
(2) No photoelectron is emitted from the cathode when V is 1.8 V.
(3) The maximum kinetic energy of the photoelectrons emitted is 1.8 eV.

A (2) only
B (3) only
C (1) and (2) only
D (1), (2) and (3)

★5 1.2 Metal plate X has a larger work function than metal plate Y. The two plates are illuminated by the same light source at the same time. Photoelectrons are emitted from both of them. Which of the following statements is correct?

A There is a shorter time delay for X to emit photoelectrons.

B There is a shorter time delay for Y to emit photoelectrons.

C If the frequency of the light source is reduced gradually, X will stop emitting photoelectrons first.

D If the frequency of the light source is reduced gradually, Y will stop emitting photoelectrons first.

★6 1.2 Metal plates X and Y are illuminated by a beam of violet light and a beam of green light respectively. The two light beams have the same intensity. Photoelectrons are emitted from both plates. Which of the following statements are correct?

(1) More photons reach plate Y than plate X per unit time per unit area.

(2) The kinetic energy of a photoelectron emitted from plate Y is always lower than that from plate X.

(3) Photoelectrons can still be emitted if the intensities of both light beams are reduced.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★7 1.2 In the photoelectric effect experiment, either the frequency or the intensity of the incident light is varied. The photoelectric current and the maximum kinetic energy of the photoelectrons emitted are recorded. A graph is then obtained (Fig b).

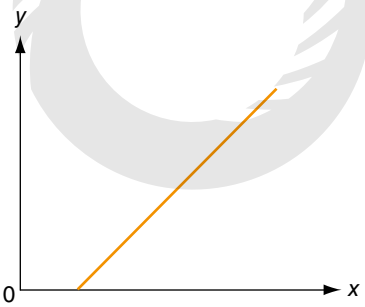


Fig b

Which of the following is/are the possible quantities represented by x and y?

	x	y
(1)	intensity of incident light	maximum KE of photoelectrons
(2)	frequency of incident light	maximum KE of photoelectrons
(3)	intensity of incident light	photoelectric current

A (2) only B (3) only

C (1) and (3) only D (2) and (3) only

★8 1.2 When a metal surface is illuminated separately by red, yellow and violet lights, photoelectrons with maximum kinetic energies K_R, K_Y and K_V are emitted respectively. If the same metal surface is illuminated by these coloured lights at the same time, what is the maximum kinetic energy of the photoelectrons emitted?

A K_R B K_V

C $K_R + K_Y + K_V$ D $\frac{1}{3}(K_R + K_Y + K_V)$

★9 1.2 A photocell is connected to the power supply as shown (Fig c).

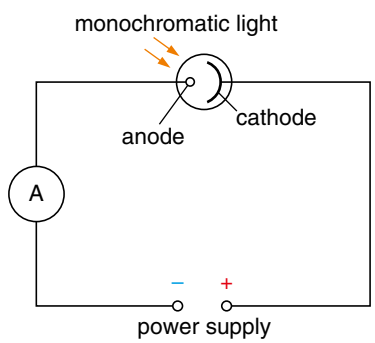


Fig c

Monochromatic light is directed onto the cathode. The frequency of the light is higher than the threshold frequency of the cathode. However, no current is detected in the circuit. Which of the following changes may produce a current?

(1) Set the e.m.f. of the power supply to zero.

(2) Use monochromatic light of a higher intensity.

(3) Use monochromatic light of a shorter wavelength.

A (1) only B (3) only

C (1) and (2) only D (1) and (3) only

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***10** 1.2 A photocell is connected to a variable power supply. Monochromatic light is incident on the cathode of the photocell so that photoelectrons are emitted. Which of the following does **not** affect the maximum kinetic energy of the photoelectrons reaching the anode of the photocell?

A The material of the cathode

B The frequency of the light

C The intensity of the light

D The output voltage of the power supply

****11** 1.2 Violet light of frequency 7.3×10^{14} Hz and intensity 0.05 W m^{-2} is directed onto the cathode of a photocell. If the area of the cathode surface receiving the violet light is 6.00×10^{-4} m^2 and one electron is emitted for every 200 photons hitting the surface, find the maximum photoelectric current.

A 4.96×10^{-8} A B 1.5×10^{-7} A

C 4.53×10^{-7} A D 9.94×10^{-6} A

Refer p.17

****12** 1.2 Figure d shows the maximum kinetic energy K_{max} of the photoelectrons emitted by metals X and Y when the plates are illuminated by monochromatic lights of different wavelengths λ.

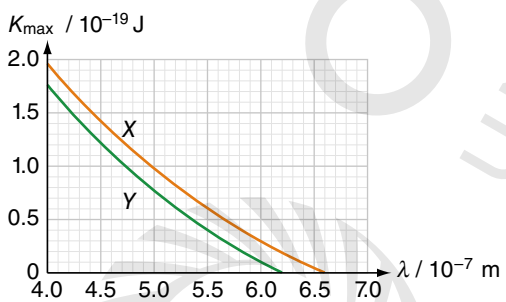


Fig d

Which of the following statements are correct?

(1) No photoelectron is emitted if monochromatic light of wavelength 7.0×10^{-7} m is shone on metal X.

(2) If monochromatic light of wavelength 5.1×10^{-7} m is shone on metal X, the stopping potential is 0.9 V.

(3) The work function of metal Y is 2 eV.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

Refer p.27, 29

13 HKDSE 2017 Paper 2 Q2.2 1.2

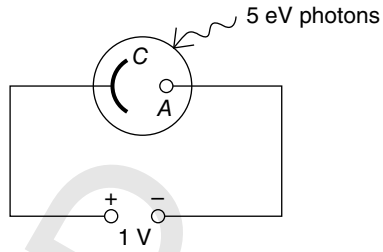


Fig e

A photocell is connected to a 1 V d.c. source as shown in Figure e. A monochromatic light beam with each photon of energy 5 eV is incident on cathode C of the photocell so that photoelectrons are emitted. If the work function of cathode C is 2 eV, what is the maximum kinetic energy of the photoelectrons reaching anode A?

A 2 eV (30%) B 3 eV

C 4 eV D 6 eV

14 HKDSE 2019 Paper 2 Q2.2 1.1

The set-up below is for investigating the maximum kinetic energy of photoelectrons in photoelectric effect (Fig f).

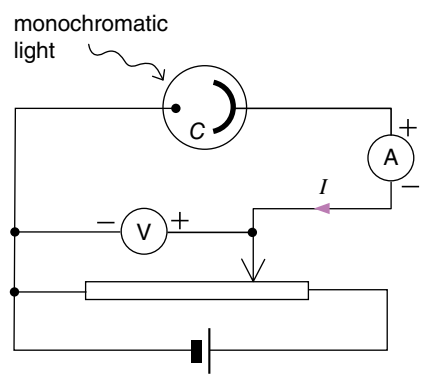
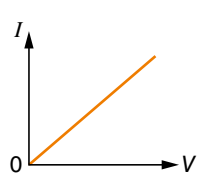
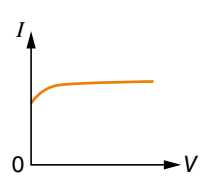
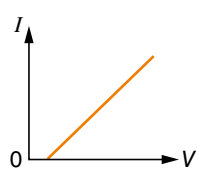


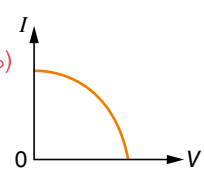
Fig f

Monochromatic light of fixed intensity is shone on the cathode C of a photocell. The p.d. V applied across the photocell is adjusted and the corresponding current I is measured. What will be the graph of I against V?

A 

B 

C 

D (25%) 

15 HKDSE 2021 Paper 2 Q2.3
1.2

When monochromatic lights of wavelength λ and 2λ are incident on a metal surface, the maximum kinetic energies of the photoelectrons emitted are in the ratio of 3:1. Find the longest wavelength of monochromatic light that can trigger photoemission for such metal.

A $\frac{5\lambda}{2}$ B 3λ

C $\frac{7\lambda}{2}$ D 4λ (39%)

Conventional questions

***16** 1.2 In the photoelectric effect experiment, a photoelectric current is detected when monochromatic light of frequency 7×10^{14} Hz is incident on the cathode. Then a p.d. is applied across the cathode and the anode and its magnitude V is increased gradually from zero. When V is 1.5 V, the photoelectric current I drops to zero.

(a) Find the maximum KE (in eV) of the photoelectrons emitted from the cathode. (1 mark)

(b) Sketch a graph to show how I varies with V from $V = 0$ to $V = 1.5$ V. (2 marks)

(c) Explain the graph in (b). (2 marks)

(d) If light of a higher intensity is used, how would the answer in (a) change? Explain. (3 marks)

(e) Find the threshold frequency of the cathode. (2 marks)

***17** 1.2 A student carries out the photoelectric effect experiment. The frequency f of the incident light and the stopping potential V_s are measured. The results obtained are shown in Table b.

f / 10^{14} Hz	V_s / V
6.9	0.76
8.2	1.06
9.1	1.69
9.9	1.91
10.6	2.11
11.8	2.44

Table b

(a) Explain what stopping potential is. (1 mark)

(b) Using the data in Table b, plot a graph of V_s against f and draw the best fit straight line. (3 marks)

(c) Estimate Planck constant from the graph. (2 marks)

(d) Using the standard value of Planck's constant,

(i) estimate the work function of the cathode in the photocell; (2 marks)

(ii) determine if photoelectrons are emitted when the cathode is illuminated by light of wavelength 500 nm. (2 marks)

(e) How will the graph in (b) change if the cathode is made of another material with a smaller work function? (2 marks)

***18** 1.2 In the photoelectric effect experiment, blue light of frequency 6.2×10^{14} Hz is directed onto the cathode. Figure g shows how the photoelectric current I varies with the p.d. V across the photocell for two different intensities of incident light. V is defined to be positive when the anode of the photocell is connected to the positive terminal of the power supply.

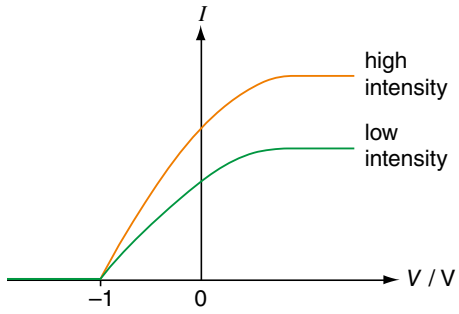


Fig g

(a) Why is I zero when $V = -1$ V? (1 mark)

(b) Explain why the two curves have the same intercept on the horizontal axis. (3 marks)

(c) Find the work function (in eV) of the cathode. (2 marks)

(d) When V increases and passes a certain positive value, I reaches a constant value. What is the physical meaning of this constant value of I? (1 mark)

(e) Find the maximum KE of the photoelectrons reaching the anode when

(i) $V = -0.5$ V, (1 mark)

(ii) $V = 2$ V. (1 mark)

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★19 1.2 A metal surface of area 5 cm × 5 cm is shone perpendicularly by green light of wavelength 550 nm. In each second, there are 1×10^{14} photons incident on the surface. The work function of the surface is 2.1 eV.

(a) Find the intensity of the green light. (2 marks)

(b) Find the maximum kinetic energy (in eV) of the photoelectrons emitted. (2 marks)

(c) The green light is replaced by red light of wavelength 680 nm.

(i) If the number of photons incident on the surface in each second remains unchanged, compare the intensity of the red light and the green light. (1 mark)

(ii) Will photoelectric effect occur? Explain. (2 marks)

★20 1.2 In the photoelectric effect experiment, EM radiation of wavelength 320 nm is directed onto the cathode (Fig h). The maximum KE of the photoelectrons emitted from the cathode is 2 eV.

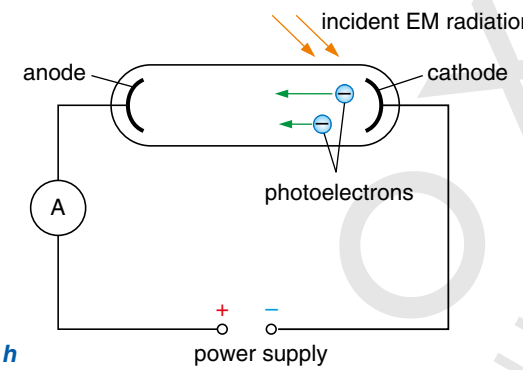


Fig h

(a) Find the energy of each incident photon (in J). (3 marks)

(b) (i) Find the threshold frequency. (2 marks)

(ii) Explain why the wave model of light fails to account for the presence of the threshold frequency. (2 marks)

(c) How does the ammeter reading change when the p.d. across the photocell increases from zero? (2 marks)

(d) How does the maximum KE of the photoelectrons reaching the anode changes when the p.d. across the photocell increases? (1 mark)

★21 1.2 A metal surface with a work function 2.9 eV is illuminated by light of photon energy 4.8 eV.

(a) State the physical meaning of work function. (1 mark)

(b) Photoelectrons are emitted immediately when light is incident on the metal surface.

(i) Explain why the wave model of light fails to predict the immediate emission of photoelectrons. (2 marks)

(ii) Find the maximum speed of the photoelectrons emitted. (1 mark)

(iii) If the intensity of light decreases, what will happen to the emission of photoelectrons? Explain. (2 marks)

(c) Will photoelectrons be emitted when the metal is illuminated by light of frequency 5×10^{14} Hz? Explain. (2 marks)

★22 1.2 In the photoelectric effect experiment, light of frequency 1.1×10^{15} Hz is incident on the cathode (Fig i). The ammeter reading is 0.5 μA. The voltmeter reading is 2 V. The threshold frequency for the cathode is 6×10^{14} Hz.

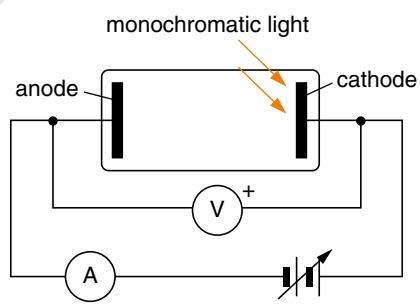


Fig i

(a) Find the maximum kinetic energy (in eV) of the photoelectrons

(i) emitted from the cathode, (2 marks)

(ii) reaching the anode. (1 mark)

(b) Find the number of photoelectrons reaching the anode per second. (2 marks)

(c) The output voltage of the power supply is varied so that the voltmeter reading decreases. How will the ammeter reading change? Explain. (2 marks)

(d) Find the stopping potential. (1 mark)

(e) Find the work function (in eV) of the cathode. (2 marks)

**23 ★★ 1.2 The surfaces of a spacecraft are made of aluminium (Fig j). When the spacecraft travels in space, the photoelectric effect occurs on the surfaces exposed to sunlight. As a result, a charge develops on the surfaces.



Fig j

(a) Which type of charge develops on the surfaces exposed to sunlight? Explain your answer. (2 marks)

(b) Aluminium has a work function of 4.12 eV.

(i) What is the physical meaning of work function? (1 mark)

(ii) Find the maximum KE (in eV) of the photoelectrons, if any, emitted from the aluminium surfaces in response to the incidence of visible light. (2 marks)

(c) Ultra-violet radiation of power 20 W in sunlight is incident on an aluminium surface. By taking the frequency of the radiation to be 1.2×10^{15} Hz, estimate the number of ultra-violet photons reaching the aluminium surface per second. (2 marks)

(d) The amount of charge built up on the aluminium surfaces by the photoelectric effect is limited. Suggest a reason. (1 mark)

Refer p.17

See p.32 for the Exam Report.

24 1.2 HKDSE 2016 Paper 2 Q.2

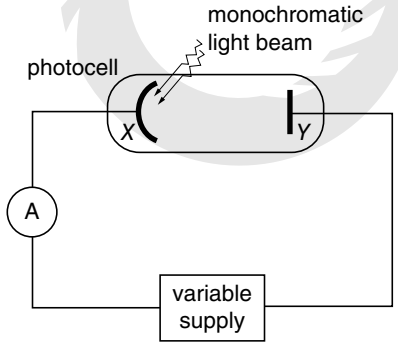


Fig k

The set-up in Figure k is for the study of the photoelectric effect. A monochromatic light beam with each photon of energy 3.4 eV is directed towards the photo-sensitive cathode X of a photocell. The potential difference V across anode Y and cathode X can be changed by adjusting the variable supply. The graph shows how the photoelectric current I varies with the potential difference V (Fig l).

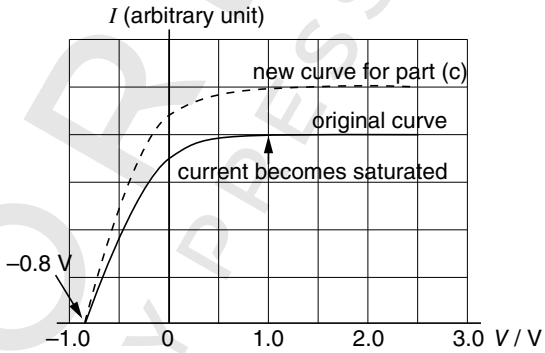


Fig l

(a) (i) The photoelectric current I becomes saturated after V reaches a certain value. Explain why this is so. (1 mark)

(ii) Hence deduce the **maximum kinetic energy**, in eV, of the photoelectrons reaching anode Y when I is just saturated. (2 marks)

(b) (i) Find the work function, in eV, of the metal of cathode X and calculate the threshold wavelength for this metal. (3 marks)

(ii) Hence explain whether yellow light of wavelength 576 nm can have photoelectric effect on cathode X. (2 marks)

(c) If the experiment is repeated with another light beam using the same photocell, a new curve (in dotted line) is obtained as shown. What can be said about this light beam's **frequency** and **intensity**? (2 marks)

See p.33 for the Exam Report.

25 1.2 HKDSE 2018 Paper 2 Q.2

The set-up shown in Figure m (p.40) can be used to measure the maximum kinetic energy of photoelectrons emitted from sodium metal when electromagnetic radiations of the same intensity but having different frequencies are incident on a sodium-coated cathode.

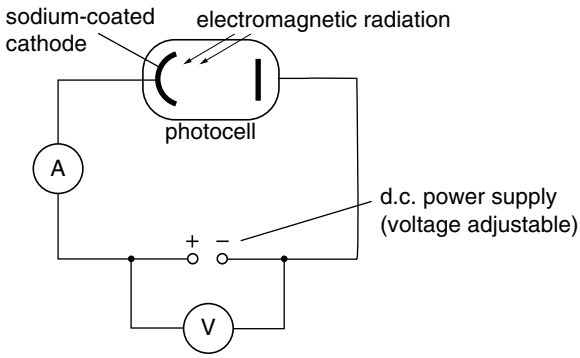


Fig m

(a) Briefly describe how the maximum kinetic energy of photoelectrons can be measured using this set-up for a certain frequency (above the threshold value) of radiation. (2 marks)

(b) The graph in Figure n shows the results of the experiment.

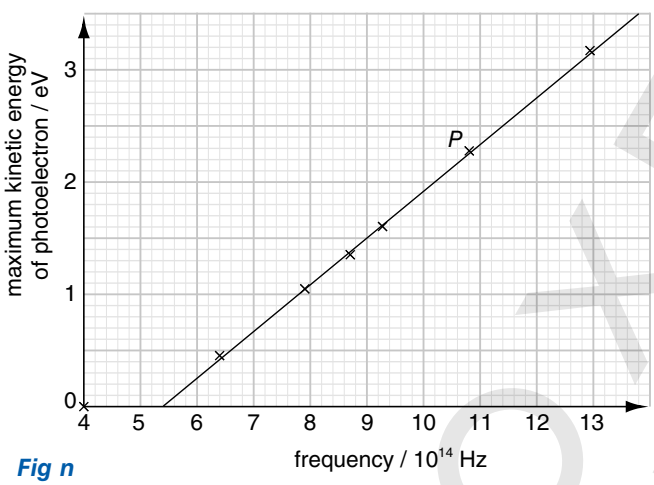


Fig n

(i) State the part of the electromagnetic spectrum from which radiation was used in obtaining data point *P*. (1 mark)

(ii) Find the slope of the graph and deduce its physical meaning. (3 marks)

(iii) Calculate the work function of sodium in eV. (2 marks)

(c) State the change, if any, in the graph obtained if the experiment is repeated with electromagnetic radiations of lower intensity. Explain. (2 marks)

See p.33 for the Exam Report.

26 **HKDSE 2020 Paper 2 Q.2**
1.2

In order to demonstrate photoelectric effect, the electrodes *A* and *C* of the photocell in Figure o are connected to a potential difference *V* which can be read from the high-resistance voltmeter. This potential difference can vary from 0 V to 2.5 V.

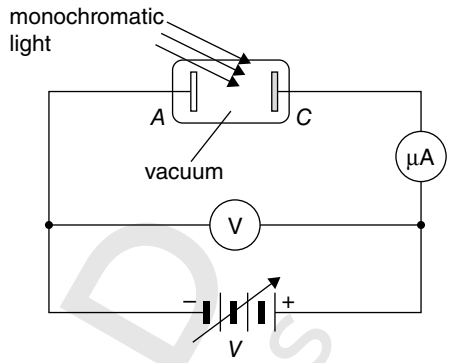


Fig o

When monochromatic light of wavelength 300 nm is incident on electrode *C*, the microammeter of negligible internal resistance shows a reading.

(a) (i) State the part of the electromagnetic spectrum (ultra-violet, blue, green, red or infra-red) that the incident light belongs to. (1 mark)

(ii) According to wave theory, there should be a 'time delay' for photoelectric emission to occur. However, the experimental result shows that photoelectric emission is immediate. State the implication of such an experimental result. (1 mark)

(b) The applied potential difference is adjusted until the microammeter reading just falls to zero when $V = 1.7$ V.

(i) State and explain whether the microammeter reading would change if an incident light of the same wavelength but higher intensity is used. (2 marks)

(ii) Calculate the work function, in eV, of electrode *C*. (3 marks)

(c) Now the applied potential difference is adjusted until $V = 0.8$ V and the microammeter registers 0.4 μA.

(i) Estimate the number of photoelectrons reaching electrode *A* in one second. (1 mark)

(ii) State the maximum kinetic energy, in eV, of the photoelectrons reaching *A*. Explain why not all photoelectrons reaching *A* possesses this amount of kinetic energy. (2 marks)